

Dysfunctional Hippocampal-Prefrontal Network Underlies a Multidimensional Neuropsychiatric Phenotype following Early-Life Seizure

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Abstract

Brain disturbances during development can have a lasting impact on neural function and behavior. Seizures during this critical period are linked to significant long-term consequences such as neurodevelopmental disorders, cognitive impairments, and psychiatric symptoms, resulting in a complex spectrum of multimorbidity. The hippocampus-prefrontal cortex (HPC-PFC) circuit emerges as a potential common link between such disorders. However, the mechanisms underlying these outcomes and how they relate to specific behavioral alterations are unclear. We hypothesized that specific dysfunctions of hippocampal-cortical communication due to early-life seizure would be associated with distinct behavioral alterations observed in adulthood. Here, we performed a multilevel study to investigate behavioral, electrophysiological, histopathological, and neurochemical long-term consequences of early-life *Status epilepticus* in male rats. We show that adult animals submitted to early-life seizure (ELS) present working memory impairments and sensorimotor disturbances, such as hyperlocomotion, poor sensorimotor gating, and sensitivity to psychostimulants despite not exhibiting neuronal loss. Surprisingly, cognitive deficits were linked to an aberrant increase in the HPC-PFC long-term potentiation (LTP) in a U-shaped manner, while sensorimotor alterations were associated with heightened neuroinflammation, as verified by glial fibrillary acidic protein (GFAP) expression, and altered dopamine neurotransmission. Furthermore, ELS rats displayed impaired HPC-PFC theta-gamma coordination and an abnormal brain state during active behavior resembling rapid eye movement (REM) sleep oscillatory dynamics. Our results point to impaired HPC-PFC functional connectivity as a possible pathophysiological mechanism by which ELS can cause cognitive deficits and psychiatric-like manifestations even without neuronal loss, bearing translational implications for understanding the spectrum of multidimensional developmental disorders linked to early-life seizures.

eLife assessment

This **important** study assesses anatomical, behavioral, physiological, and neurochemical effects of early-life seizures in rats, describing a striking astrogliosis and deficits in cognition and electrophysiological parameters. The **convincing** aspects of the paper are the wide range of convergent techniques used to understand the effects of early-life seizures on behavior as well as hippocampal prefrontal cortical dynamics. While reviewers thought that the scope was impressive, there was criticism of the statistical robustness and number of animals used per study arm, as well as the lack of causal manipulations to determine cause-and-effect relationships. This paper will be of interest to neurobiologists, epileptologists, and behavioral scientists.

Introduction

The early years of life represent a critical period for neuronal maturation and the development of cognitive functions (Short and Baram, 2019 [↗](#)). Disturbances during this period can have a lasting impact on neuronal activity and cause multifaceted neuropsychiatric manifestations (Marín, 2016 [↗](#)). Seizures are a prevalent disturbance in the immature brain and are often associated with profound and enduring adverse neurological outcomes that persist throughout adulthood (Brunquell et al., 2002 [↗](#); Holmes, 2009 [↗](#)). Children who experience early-life seizures (ELS) not only face an increased likelihood of developing epilepsy but also have a heightened susceptibility to cognitive impairments and behavioral disorders, including autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), intellectual disability, and psychiatric symptoms like schizophrenia (Dreier et al., 2019 [↗](#); Holmes, 2009 [↗](#); Kaufmann et al., 2009 [↗](#)). Since the pathophysiological mechanisms linking ELS and neurodevelopmental disorders are complex, understanding how ELS *per se* affects neuronal function and dissecting the associations between neurobiological and behavioral alterations are crucial for developing better treatments for these conditions.

In experimental models, ELS induces long-term cognitive and behavioral alterations with translational validity with the main neuropsychiatric domains affected in humans (Holmes, 2016 [↗](#)). Studies have reported that a prolonged ELS in rodents can lead to cognitive deficits, increased anxiety, reduced social interaction, and facilitated development of epilepsy (Castelhano et al., 2013 [↗](#); De Oliveira et al., 2008a [↗](#); Kubova et al., 2004 [↗](#)). Interestingly, ELS models do not exhibit severe neuronal loss or structural damage (Baram et al., 2002 [↗](#); Sinh et al., 2003 [↗](#)), suggesting that the behavioral alterations induced by ELS may be mediated by functional and molecular effects at the brain circuit level (Kandratavicius et al., 2012b [↗](#), 2012a [↗](#)). The hippocampal-prefrontal cortical circuit (HPC-PFC) plays a crucial role in cognitive and sensorimotor processes, and evidence converges to indicate its dysfunction as a possible common mechanism shared between ELS, neurodevelopmental, and psychiatric disorders, especially epilepsy, psychosis/schizophrenia, ASD, and cognitive disability (Ruggiero et al., 2021 [↗](#); Sigurdsson and Duvarci, 2016 [↗](#)). Reports indicate that ELS impacts HPC-PFC functional connectivity, as revealed by the coordination of network oscillations during particular situations (Barry et al., 2020 [↗](#); Kleen et al., 2011 [↗](#); Mouchati et al., 2019 [↗](#); Niedecker et al., 2021 [↗](#)). However, how specific functional or molecular effects of ELS on the HPC-PFC network are associated with separate behavioral dimensions within this complex multimorbidity spectrum remains unknown. Also, since behavioral alterations of ELS are long-lasting and progressive, a comprehensive characterization of network dynamics across brain states would be pivotal for a complete understanding of the effects of ELS on brain function.

Here, we hypothesized that ELS would produce long-term alterations in HPC-PFC neuronal dynamics related to behavioral abnormalities (Ruggiero et al., 2012 [↗](#)). To test this hypothesis, we induced *Status epilepticus* (SE) as a model of prolonged ELS and performed behavioral tests to assess cognitive and sensorimotor processes. Then, we performed neurophysiological, immunohistochemical, and neurochemical experiments to study potential changes in the HPC-PFC circuits that could underlie such behavioral alterations. We found that adult ELS rats present a myriad of behavioral impairments: working memory deficits, hyperlocomotion, poor sensorimotor gating, and behavioral sensitivity to psychostimulants. Surprisingly, we found an enhanced long-term potentiation (LTP) in the HPC-PFC pathway of ELS animals correlated with cognitive impairment in a U-shaped manner. Our results also describe neuroinflammation and alterations in dopaminergic neurotransmission as significant contributors to sensorimotor disturbances. We also investigated HPC-PFC dynamics across different brain states in freely-moving rats. ELS rats displayed an increased HPC theta power and impaired coordination of PFC gamma activity during active behavior. During rapid eye movement (REM) sleep, ELS rats showed a reduction in HPC-PFC coherence. Remarkably, ELS rats presented an aberrant brain state during active behavior characterized by oscillatory dynamics oddly similar to REM sleep. Finally, we implemented a linear discriminant model, demonstrating that collective patterns of oscillatory alterations accurately distinguished ELS from control animals. Interestingly, although these alterations relate to broad clinical manifestations of neurodevelopmental disturbances and psychiatric manifestations, we recognize a particular convergence of dysfunctional patterns at multiple levels linking the consequences of ELS to functional alterations reported in animal models of epilepsy, psychosis/schizophrenia, ASD, and cognitive disability. These convergent patterns indicate that epileptogenic processes triggered during early life, such as neuroinflammation, could lead to altered neurotransmission and, consequently, aberrant dynamics in the HPC-PFC coordination, potentially contributing to the neurological and psychiatric manifestations linked to early-life seizure.

Materials and Methods

Animals

Wistar rats were housed in a colony room with controlled temperature ($22 \pm 2^\circ\text{C}$) and a 12-h light/dark cycle with free access to food and water. Litters contained up to 8 animals with a maximum of 5 males. Rats were weaned at 21 days postnatally (P21). All procedures were performed according to the National Council for the Control of Animal Experimentation guidelines for animal research and approved by the local Committee on Ethics in the Use of Animals (Ribeirão Preto Medical School, University of São Paulo; protocol: 159/2014).

Experimental Designs

We used four cohorts (a total of 17 litters) to compare the effects of early-life *Status Epilepticus* (ELS) versus control (CTRL) animals. The first cohort was used for acute electrophysiology during the induction of ELS (n=5, ELS only). The second was used for behavioral tests (CTRL n=11, and ELS n=14), synaptic plasticity experiments (CTRL n=7, and ELS n=7), and immunohistochemistry (CTRL n=9, and ELS n=11). The third was used to investigate HPC-PFC dynamics in freely moving rats (CTRL n=6, and ELS n=9). Finally, in the fourth cohort, we assessed behavioral sensitization to psychostimulants (CTRL n=15, and ELS n=14) and neurochemical quantification (CTRL n=8, and ELS n=9).

Early-Life Status Epilepticus

At P11-P12 we induced a 2-hour SE using lithium chloride (LiCl, 127 mg/kg, i.p., Sigma-Aldrich, USA) 18-20 hours prior to injection of methylscopolamine (1 mg/kg, i.p., Sigma-Aldrich) and pilocarpine (80 mg/kg, s.c., Sigma-Aldrich, USA) (Kubová and Mareš, 2013 [↗](#)). Behavioral seizures

were monitored following the Racine scale (Racine, 1971 [↗](#)). SE was interrupted by diazepam (5 mg/kg, i.p., Teuto, Brazil) at the 2-hour mark. During the recovery period (1 week post-ELS) we provided food supplementation and subcutaneous rehydration with saline and monitored body temperature and weight.

Behavior

The radial-arm maze working memory task was based on Floresco et al. (1997) [↗](#). Briefly, rats were kept at 85-90% of their initial body weight throughout the 21 days of the experiment. Training consisted of single daily sessions, each one divided into the training phase and test phase, 30 min apart. During the training session, four doors were kept open providing access to food pellets. In the test session, all doors remained open, but with pellets available only in the four arms that were closed in the previous training session. The sessions were terminated upon full pellet consumption or after 5 min. Two types of errors were quantified: Reference error: reentering arms previously visited during the training session; Working memory error: reentering an arm already visited during the test session. The sum of both categories denoted the total errors for each animal.

Three days after returning to *ad libitum* diet, locomotor activity was evaluated for 30 min in the open field test (Wolf et al., 2016 [↗](#)).

Immediately after the open field test, rats were examined for prepulse inhibition (PPI) of the acoustic startle. The PPI test consisted of a habituation period (5 min) followed by 72 trials (intertrial interval: 15 ± 8 s) with prepulse alone (PP), startling pulse alone (P), prepulse followed by a startling pulse (PP + P), and no stimulus, all of which accompanied by background noise (65 dB white noise). Startle-eliciting stimuli were presented at 120 dB for 40 ms (P) and prepulse stimuli were presented for 20 ms (PP, 1000 Hz; 71, 77, and 83 dB; 20 ms). Trials were presented in a random order, with each trial type being presented six times. We used the equation: $\%PPI = 100 - 100 \times (\text{mean startle amplitude at PP} + P) / (\text{mean startle amplitude at P})$ (Wolf et al., 2016 [↗](#)).

In order to measure the effects of ELS on sensitization to psychostimulants, we performed the apomorphine-induced locomotion test. At P80-90, rats were placed in the open field apparatus and were allowed to roam for 60 min. Then, we injected apomorphine (APO, 1.5 mg/kg, i.p., Sigma-Aldrich, USA) and measured locomotion for another 60 min. The distance across blocks was normalized by Z-score to the mean of baseline blocks.

Synaptic Plasticity

HPC-PFC synaptic plasticity studies were performed as described in previous works from our group (Bueno-Junior et al., 2018 [↗](#); Ruggiero et al., 2018 [↗](#)). Briefly, rats were anesthetized with urethane (1.2 mg/Kg in NaCl 0.15 M, i.p.) for electrode implantation. A recording electrode (Teflon-coated tungsten wires 60 μ m, AM-Systems) was inserted in the medial PFC (prelimbic region (PL), anterior-posterior: 3.0 mm; medial-lateral, ML: 0.5 mm; dorsal-ventral, DV: 3.2 mm), and a bipolar stimulation electrode (twisted wires; ~500 μ m inter-pole distance) was inserted in the ipsilateral HPC (intermediate CA1 AP: -5.6 mm; ML: 4.5 mm; DV: 2.5 mm). An additional burr hole was drilled over the right parietal cortex for a reference screw. Temperature was kept constant at $37 \pm 0.5^\circ\text{C}$ during the experiment.

HPC was stimulated with single monophasic pulses (200 μ s, 0.05 Hz, 150–300 μ A, 80-ms inter-pulse interval) while recording PFC field postsynaptic potentials (fPSPs). Stimulation intensity (60–500 μ A) was calibrated to evoke 50–60% of the maximal fPSP amplitude. Field recordings were conditioned through a preamplifier (100 $\times$ gain, 0.03–3 kHz bandpass; Grass) and digitized at 10 kHz (ADInstruments). For LTP induction, we used a high-frequency stimulation (HFS) protocol consisting of 2 series (10 min apart) of 10 trains (0.1 Hz), each train with 50 pulses at 250 Hz (Ruggiero et al., 2018 [↗](#)). fPSP amplitudes and slopes were computed as the difference and slope

between the positive and negative peaks, respectively. Short-term synaptic plasticity was estimated by paired-pulse facilitation (PPF), calculated as the ratio of the fPSP2 and fPSP1 (Lopes-Aguiar et al., 2020 [DOI](#)). Amplitudes and slopes were normalized as ratios of the baseline mean and averaged into 10-min bins.

Brain Tissue Processing

Immunohistochemistry

We employed a standardized immunohistochemistry protocol (Peixoto-Santos et al., 2015 [DOI](#)) to estimate NeuN⁺ neuronal density and to evaluate parvalbumin-expressing (PV) interneurons, mGluR5 expression, and glial fibrillary acid protein (GFAP) reaction. Briefly, endogenous peroxidases were blocked and the antigens were exposed by microwave-induced retrieval. We used the primary antibodies against NeuN (Chemicon, USA; 1:1000), PV (Calbiochem, USA; 1:500), mGluR5 (Millipore, USA; 1:200), and GFAP (Dako, Denmark; 1:500) proteins, and biotinylated secondary antibodies (Dako, Denmark; 1:200). Revelation was performed with Elite ABC Kit (Vector) and diaminobenzidine solution (Pierce).

Images from the regions of interest were captured at 200× magnification in the Axio Scope.A1 microscope system (Carl Zeiss). The regions of interest included the hippocampus (CA1 and GL, granule layer of dentate gyrus), entorhinal cortex (EC), infralimbic (IL) and prelimbic cortex (PL), and thalamic reticular nucleus (TRN). Images were then processed in Image J software (NIH, USA; 1.48v). Data are shown as the density of NeuN positive cells and immunopositive area for PV, mGluR5, and GFAP.

High-Performance Liquid Chromatography (HPLC)

Samples were taken bilaterally from the nucleus accumbens (NAc), PFC, and HPC. The samples were homogenized with ultrasound in a 0.1 mol/L HClO₄ 0.02% Na₂S₂O₂ solution containing dihydroxybenzylamine (DHBA, 146.5 ng/mL), and homoserine (HSER, 10 µg/mL) as internal standards and at a dilution of 15 µL per sample milligram. We employed Shimadzu LC-10AD isocratic systems with a fluorescence detector for glutamate (Glu) and gamma-aminobutyric acid (GABA), and an electrochemical detector for dopamine (DA) and its metabolites 3,4-dihydroxyphenylacetic acid (DOPAC) as previously published (Castro-Neto et al., 2013 [DOI](#)). The neurotransmitter levels were considered to reflect neurotransmitter stock in synaptic vesicles, their metabolites reflected the release, and metabolite/neurotransmitter ratios were taken as an index of neurotransmitter turnover. Standards containing all amino acids and monoamines/metabolites of interest were employed to define their respective peaks in the samples.

Freely Moving Electrophysiology

Electrophysiological recordings

Chronic head caps consisted of two eight-channel connectors (Omnetics), one for the PFC (PL, 7 wires) and another for the HPC (intermediate CA1, 7 wires) (Bueno-Junior et al., 2017 [DOI](#); Marques et al., 2022 [DOI](#)). We additionally implanted a recording wire into the trapezius muscle to serve as an electromyogram (EMG). The EMG wire was soldered to the HPC connector. Microscrews were fastened into the skull, including a ground reference in the intraparietal area. Animals were allowed to recover for 8-9 days before recordings.

To record electrophysiological signals through sleep-wake states, rats were placed into a shuttle box apparatus inside a soundproof box with food and water *ad libitum*. Recordings varied between 30 to 48 hours, divided later into periods of 6 hours for analysis. Local field potentials

(LFP) were recorded through a multichannel acquisition processor (Plexon) (1000× gain, 0.7-500 Hz bandpass filter, 1-kHz sampling rate).

Classification of sleep-wake states

Classification of the sleep-wake cycle states was carried out through a semi-supervised machine learning pipeline. The HPC theta (4.5-12 Hz), PFC delta (0.7-4 Hz), and EMG (100-200 Hz) power were computed in 5-second epochs without overlaps (4 tapers; time-bandwidth product: 2.5). A support vector machine (SVM) (radial basis function) was trained to discriminate the epochs into: (1) awake or sleep epochs using EMG power; (2) REM or NREM epochs based on theta/delta ratio; (3) quiet wake (QWK) or active (ACT) epochs based on theta/delta ratio. Later, we classified our recordings based on 30-second epochs by grouping sets of six 5-second epochs and considering the most representative state.

Spike wave-discharge detection (SWD)

For spike wave-discharge detection, we used the PFC recordings. First, the short-time Fourier transform (STFT) was computed within 3-70 Hz throughout 250-ms windows with 50% overlap. A detection threshold was defined as the mean + 1.5 standard deviation of the absolute value of the STFT of the frequency bands 4.4-16.4, 17.6-32.8, and 35.1-55 Hz throughout the entire recording. SWD events were detected when the average STFT of the aforementioned frequency bands at one epoch exceeded the threshold. Consecutive epochs above the threshold were grouped as the same event. The beginning and the end of an event were defined as the closest epochs when the average STFT was lower than the mean + 1 standard deviation of STFT of the frequency bands of interest. Lastly, the detected putative events were manually sorted by an expert.

Spectral estimates

We divided 30-s epochs into 3-s epochs for spectral analyses. Power spectrum densities (PSD) and magnitude-squared coherence were obtained using Welch's method with 1-s segments, 50% overlap, and Hamming windowing. Relative power was obtained by dividing values by the sum within 0.5-55 Hz for each epoch. We defined the frequency bands: delta (1-4 Hz), theta (4.5-11 Hz), low gamma (30-55 Hz), and high-gamma (65-100 Hz) to compare estimates and obtained the spectral peak of each epoch using MATLAB function *findpeaks*.

To quantify theta oscillation power we used the FOOOF algorithm (Donoghue et al., 2020 [DOI](#)), which parametrizes the PSD separating the periodic and aperiodic components of the signal. We used frequencies within 3-40 Hz, the “fixed” aperiodic mode, a minimum peak height of 0.05, and a maximum number of peaks of 4 (<https://fooof-tools.github.io/fooof/> [DOI](#)). EMG data were low-cut filtered (>30 Hz) and the sum of power within 30-200 Hz was obtained.

As a synchrony measure, we also computed cross-correlations between each pair of LFPs (low-pass filter of 50 Hz) recorded from the HPC and PFC using a maximum lag of 1 s. Coefficients were normalized so that the autocorrelations at zero lag were one. We observed that the highest coefficients occurred in the range of -80 to -20 ms (HPC leading the PFC), so we obtained the maximum coefficient within this range for each rat average for comparisons. Cross-spectrum power was calculated using Welch's method for the normalized correlation coefficients from -1 to 1 s.

For cross-frequency amplitude spectrum correlations, we computed the absolute STFT for each 3-s epoch throughout 100-ms segments with 50% overlap and Hamming windowing across 0.5-Hz frequency bins within 0.5-200 Hz. We calculated the correlation matrix for each frequency pair using Pearson's R.

State maps

After state classification and spectral analysis, we computed two-dimensional state maps to investigate the natural distinction between sleep-wake states. The state maps were computed according to Gervasoni et al. (2004) [↗](#) and Dzirasa et al. (2006) [↗](#). First, we obtained the power *ratio 1* (4.5-9/4.5-50 Hz) and *ratio 2* (2-20/2-50 Hz) for each region and the first principal component, estimated by singular value decomposition algorithm considering both regions, for each ratio. Finally, we applied a Hann window of 10 epochs on scores throughout time for smoothing.

To identify the intermediate states (IS), we first estimated the trajectory *speed* of epochs throughout state map dimensions as the Euclidean distance between each epoch and its predecessor. Then, we defined a threshold of 0.1 and labeled the epochs with greater speed as IS. This threshold was determined heuristically, corresponding closely to the expected value plus a standard deviation from a Poisson distribution across rats.

From visual inspection and the literature (Dzirasa et al., 2006 [↗](#); Gervasoni et al., 2004 [↗](#)), the state maps usually separate three well-defined clusters heuristically corresponding to awake, REM, and NREM sleep. To assure the correspondence between state map clusters and the classification outcomes, we performed Gaussian mixture models clustering of the state maps for 3 clusters and 80% posterior probabilities and computed Chi-squared tests with the classified NREM, REM, and awake (QWK+ACT) labels. Three animals were excluded for having outlier lower Chi-values. Although QWK and ACT states are often evenly distributed inside the awake cluster, QWK epochs may overlap with intermediate states closer to NREM, so the ACT epochs were considered to represent more precisely the localization of neural dynamics associated with behavior in the state map. The dissimilarity between ACT and REM states was estimated by the average pairwise state map Euclidean distance of all epochs between the two states.

We used machine learning algorithms to decode sleep-wake states from the state map. We used the ACT and REM state map *ratios 1* and *2* values. We applied a linear and radial basis function SVM *fitcsvm* function in MATLAB ('KernelFunction', 'linear' or 'rbf') as well as a Random Forest algorithm using the *fitcensemble* function in MATLAB ('Method', 'bag', 'NumLearningCycles', 50, for Random Forest) to decode states. We used the ACT and REM epochs from all the rats creating one single model for the CTRL or ELS condition. Using the same approach, we also constructed a random forest decoder model with non-reduced variables using 1 to 55 Hz (0.5 Hz bins) for the PFC and HPC power and coherence. Discrimination was evaluated using the area under the receiver operating characteristic curve (AUC-ROC). The point-wise confidence interval was calculated using a 1000-repetition bootstrap. Average AUC-ROC values were calculated using the prediction of the trained model on 20% of the epochs as test data and repeating the process 100 times.

Statistical Analysis

Data were analyzed using GraphPad Prism Software version 9.00 (GraphPad Software, LLC), and R studio version 2021. We used paired Student *t*-tests for within-group comparisons and two-sample *t*-tests for between-group comparisons. Two-way ANOVA with repeated measures (RM) was performed for comparisons across time bins, using Sidak's multiple comparison test as a *post-hoc*. To examine linear model assumptions, we analyzed the distribution of the residuals using a Q-Q plot, the Kolmogorov-Smirnov test, homoscedasticity, and the residual plot. We used the Mann-Whitney test to compare two groups and the Wilcoxon signed-rank test to paired comparisons as non-parametric alternatives. We calculated Pearson's linear correlation and Spearman's rank correlation as a nonparametric equivalent. To compare correlation and spectral coherence

estimates we applied Fisher's Z transform to the data. For multiple comparisons in the neurochemical quantification, we performed Holm's procedure for multiple comparisons correction across neurotransmitters and metabolites per region.

Data are expressed as the mean $\pm$ SEM. The significance level was set to 0.05 unless stated otherwise. For some multiple comparisons, statistical values are expressed as the range. Significances are expressed as $^{\dagger}p < 0.1$, $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$.

Linear mixed-effects models

We also constructed statistical inferential models across time controlling for individual variability and litter using linear mixed-effects models. We compared power and coherence estimates throughout time by dividing the electrophysiological recording into 5 periods (6 hours). Group and time (periods) were assigned as fixed effects and subject, litter, and time were assigned as random effects. We used this approach to construct a maximal random effects structure model as preconized in the literature as more generalizable (Barr et al., 2013 [↗](#)). The model was estimated by restricted maximum likelihood using the *lme* function of the *nlme* package (Pinheiro et al., 2023) with an unstructured covariance matrix.

Multivariate analysis

Each variable was Z-scored before multivariate analysis. We constructed correlograms to assess the relationship within behavioral and neurobiological measures. Variables were sorted by agglomerative hierarchical clustering using Ward's method and correlation as the metric. We also performed agglomerative hierarchical clustering of subjects using Ward's method and Euclidean distance as the metric to assess how unsupervised clustering by our multivariate data might match the experimental groups. An optimal number of clusters was estimated using the silhouette method and final definition was based on parsimonious interpretability of the results.

Principal component analysis (PCA), using singular-value decomposition, was used for dimensionality reduction to find collective patterns of variance within behavioral variables (behavior factors) and neurobiological variables (neurobiology factors), and to reduce each immunohistochemical measure across all brain regions into single scores. Principal components (PCs) were projected onto data and the mean score was compared between conditions.

Canonical correlation analysis (CCA) was performed to describe multivariate relationships between behavioral and neurobiological measures. Missing data were imputed using the multiple imputation by chained equations algorithm. For CCA, we used the first PC (PC1) and second PC (PC2) of behavioral variables as inputs for the behavioral factors and the PC1 of each immunohistochemical variable (NeuN, PV, mGluR5, and GFAP) across all brain regions and the Z-scored LTP data as inputs for the neurobiological factors. Variables were narrowed down from 30 to 7, which is statistically appropriate for our sample size. We used the CCA package in R to obtain the canonical dimensions and performed the Hotelling-Lawley trace statistic to assess the significance of each dimension.

We used multivariate generalized linear models (GLM) with a Gaussian family and an identity link function. Quasi-Poisson regression (quasi-Poisson family and logarithmic link function) were used for count data with overdispersion (Errors in the radial maze) using a quadratic covariate (LTP). We constructed two types of models: 1) a full model considering all the covariates; and 2) a restricted model using backward stepwise model selection based on the Akaike information criterion.

To assess the effectiveness of neurophysiological features in discriminating CTRL from ELS rats, we fitted a regularized linear discriminant classifier model on the PCA scores of selected electrophysiological variables. We used the MATLAB function *fitcdiscr* without hyperparameter

optimization. The linear classifier using two predictors allowed the graphical representation and avoided the overfitting of a high number of variables. Accuracy was determined as the probability of correct category assignments using leave-one-out cross-validation. We also determined the contribution of each original neurophysiological variable by assessing their respective AUCs using a univariate logistic regression model.

Results

ELS induces cognitive and behavioral impairments in adulthood

SE induced by lithium-pilocarpine in early life (**Fig. 1A**) produced an unequivocal and stereotyped seizure, consisting mainly of forelimb myoclonus and orofacial automatism (Racine 3), with some animals showing further development into bilateral forelimb myoclonus, rearing and falling (Racine 4 and 5) (Racine, 1971). Seizures were confirmed electrophysiologically in freely moving recordings of the frontal cortex and HPC, with paroxysms starting 10 min after pilocarpine injection and persisting for at least 120 min (**Fig. 1B**).

Next, we investigated the long-term behavioral consequences of ELS (**Fig. 2A**). First, we evaluated possible weight differences as a confounder of behavioral measures. ELS rats displayed weight loss on the day after SE (Two-way RM ANOVA, treatment effect: $F_{(9,207)} = 11.7$, $p < 0.0001$, Sidak's *post-hoc*: $p < 0.0001$; **Fig. 2B**). However, ELS animals recovered the pace of normal weight gain in the subsequent days (*post-hoc*: $p > 0.05$) and into adulthood (**Fig. 2C**), showing no distinction to CTRL before (*t*-test, $t_{(23)} = 1.440$, $p = 0.1634$) or during food restriction (*t*-test, $t_{(23)} = 0.0535$, $p = 0.9577$), indicating that behavioral effects were unrelated to weight differences or alterations in food-seeking.

Importantly, when examined for spatial memory in the radial-arm maze, ELS rats presented a specific increase in working memory errors (Mann-Whitney test, $U = 36.5$, $n(\text{CTRL}, \text{ELS}) = 11, 14$, $p = 0.0254$; **Fig. 2D**). We found no differences in the total reference errors (*t*-test, $t_{(23)} = 1.719$, $p = 0.0991$; **Fig. 2D**) and in the learning curve (Two-way RM ANOVA, treatment effect: $F_{(1,23)} = 0.8032$, $p = 0.3794$; **Fig. 2E**). In addition, ELS rats exhibited significantly higher locomotion in the open field test (Mann-Whitney test, $U = 27$, $n(\text{CTRL}, \text{ELS}) = 11/14$, $p = 0.0051$; **Fig. 2F**). ELS rats also displayed a higher startle response (*t*-test, $t_{(23)} = 2.757$, $p = 0.011$; **Fig. 2G**) and a deficit in sensorimotor gating represented by a reduction in PPI of the acoustic startle (Two-way RM ANOVA, treatment effect: $F_{(2,46)} = 19.6$, $p < 0.0001$, Sidak's *post-hoc*: PPI 71 dB: $p = 0.0017$; PPI 77 dB: $p = 0.1040$; PPI 83 dB: $p = 0.030$; **Fig. 2H**).

Interestingly, all these behavioral alterations are described in animal models of neurodevelopmental disorders such as ASD and schizophrenia (Ey et al., 2011; Scott et al., 2018; Sigurdsson, 2016). We investigated if collective patterns of behavioral alterations induced by ELS could reflect latent factors representing more comprehensive behavioral dimensions. Working memory errors in the radial-arm maze showed little correlation with the other behavioral variables, while sensorimotor variables (locomotion, startle response and sensorimotor gating) presented a strong correlation with each other. PPI exhibited a positive correlation with every intensity and a negative correlation with startle response and locomotion (**Fig. 2I**). We also performed PCA to extract patterns of variance between multivariate behavioral data (**Fig. 2J-M**). Our analysis indicated that dimensionality reduction by PCA using the first two components explained 71.32% of the variability of the data (**Fig. 2J**). By analyzing the loadings of the PC1, the variables that contribute more strongly (>0.5) to this dimension were: high total distance, startle reflex, and low PPI scores, indicating this component represents a latent factor of sensorimotor dysfunction. In turn, the PC2 represents the cognitive dimension of the behavioral variables, as it presents a high loading value (>0.5) for working memory errors (**Fig. 2K**). Finally, we found that these PCs clearly distinguished CTRL from ELS animals, which

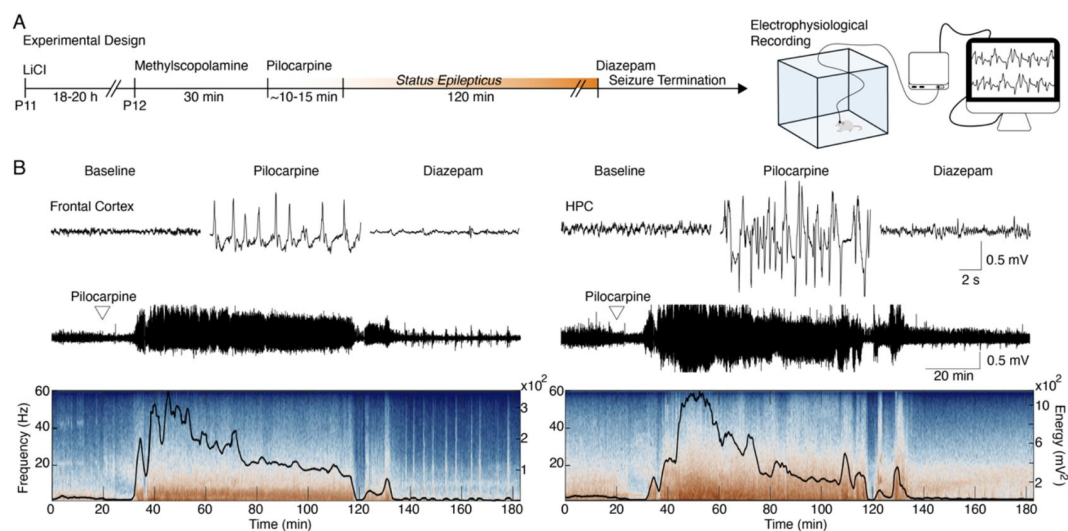


Figure 1.

ELS induction.

A, Experimental design representing early-life SE induction and monitoring. **B**, Representative traces of LFP during baseline, motor seizure induced by pilocarpine, and ELS interruption with diazepam injection (top). Representative LFP from the frontal cortex and HPC throughout the experiment (middle). LFPs are aligned with their respective spectrograms and energy curves (bottom).

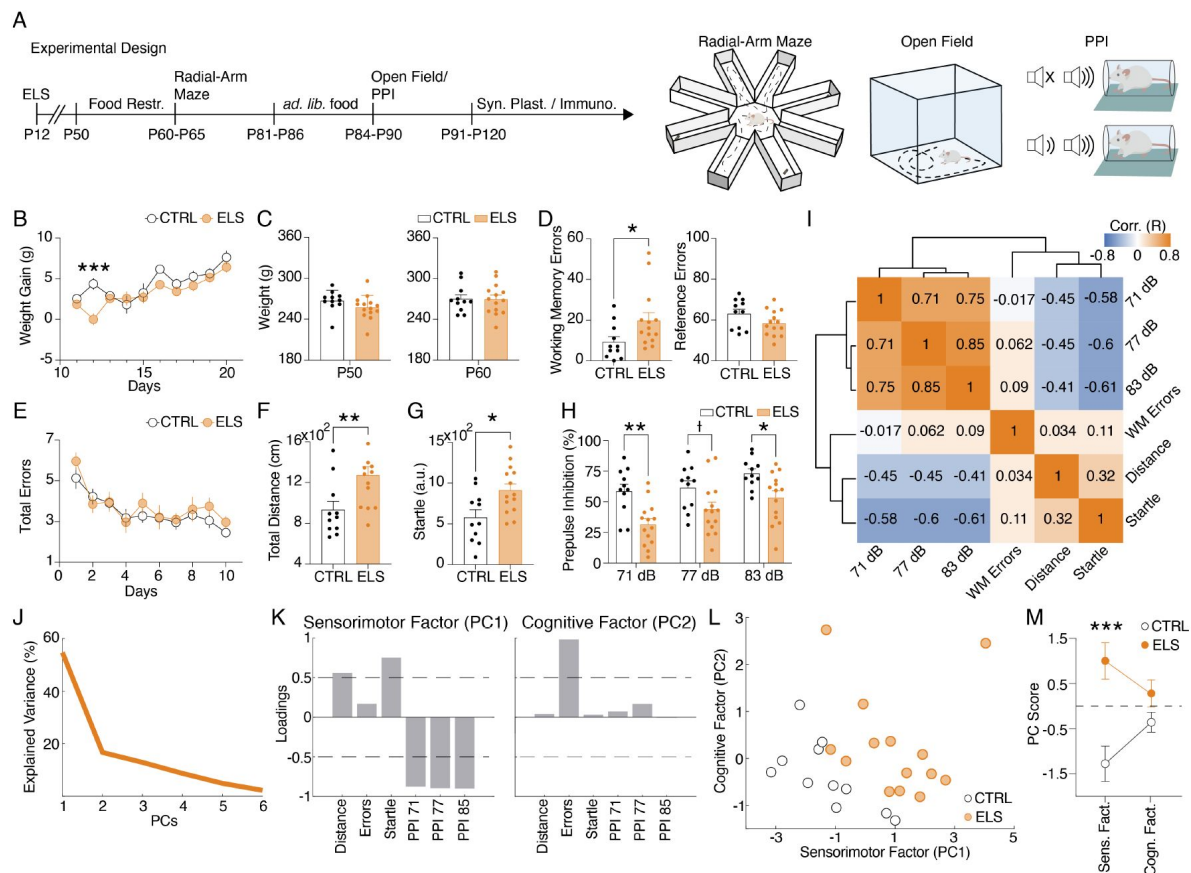


Figure 2.

ELS induces cognitive and sensorimotor impairments.

A, Experimental design representing early-life SE induction, age of the animals during the experiments, and behavioral tests performed. **B**, Weight gain after ELS induction. ELS rats showed weight loss only on the ELS induction day. **C**, No weight differences between ELS and CTRL throughout experiments in adulthood, regardless if before food restriction (left) or during food restriction (right). **D**, Working memory task on the radial-arm maze. **E**, ELS rats showed an increase in working memory errors across sessions (left) without differences in the number of reference errors (right). **F**, Hyperlocomotion of ELS rats in the open field test. **G**, Higher acoustic startle of ELS rats. **H**, Impaired sensorimotor gating in ELS rats as measured in the PPI test with three stimulus intensities. **I**, Correlation matrix of all behavioral variables. Note a cluster of positive correlations formed by the PPI measures and another cluster of negative correlations formed by the PPI measures with distance traveled on the open field and the acoustic startle. Working memory errors were not correlated with the other behavioral features. **J**, Explained variance from principal components analysis (PCA). Note that the two first PCs explain 71.32% of the data. **K**, Loadings of the behavioral variables on each principal component (PC) reveal a *sensorimotor factor* (PC1) and an uncorrelated *cognitive factor* (PC2). **L**, PCA of the behavioral features showing each animal projected onto the new factors. **M**, The sensorimotor factor distinguishes ELS from CTRL animals, as we compare the PC scores from each group. † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Error bars represent the mean $\pm$ SEM.

presented higher values of PC1 (PC1: t -test, $t_{(23)} = 3.9713$, $p = 0.0006$; PC2: t -test, $t_{(23)} = 1.6466$, $p = 0.1132$; **Fig. 2L-M**). These results indicate that the covariance across behavioral tests can be summarized by a latent *sensorimotor factor* (PC1) and a *cognitive factor* (PC2), representing two distinct dimensions of the behavioral phenotype and suggesting possibly different neurobiological bases.

Aberrant HPC-PFC synaptic plasticity in animals submitted to ELS

Next, we tested the hypothesis that behavioral abnormalities in ELS rats could be associated with synaptic plasticity impairments of the HPC-PFC pathway (**Fig. 3A-C**). First, we looked for ELS-related alterations in basal synaptic efficacy before inducing LTP. Synaptic efficacy was assessed in terms of basal fPSP amplitude and slope (amplitude: t -test, $t_{(12)} = 1.278$, $p = 0.2255$; slope: t -test, $t_{(12)} = 0.5470$, $p = 0.5944$; **Fig. 3D**), input-output curve (fPSP amplitude as a function of stimulus intensity) (Two-way RM ANOVA, treatment effect: $F_{(1,12)} = 0.6596$, $p = 0.4325$; **Fig. 3E**) and PPF curve (PPF as a function of the inter-pulse interval) (Two-way RM ANOVA, treatment effect: $F_{(1,12)} = 0.3345$, $p = 0.5737$; **Fig. 3F**). None of these measures indicated alterations in ELS when compared to CTRL (**Fig. 3D-F**).

Then, we proceeded to the synaptic plasticity experiments and found that ELS rats presented an abnormally augmented LTP compared to CTRL (**Fig. 3G**; Two-way RM ANOVA, treatment effect: $F_{(1,12)} = 7.54$, $p = 0.0177$; interaction effect: $F_{(26,312)} = 7.393$, $p < 0.0001$; Sidak's *post-hoc*: $p < 0.05$ for 10-60 min; t -test, $t_{(12)} = 2.747$, $p = 0.0177$; **Fig. 3G** inset), despite the lack of alterations in basal synaptic efficacy described above (**Fig. 3D-F**). Aberrant LTP in the ELS rats was also observed when analyzing the fPSP slope (Two-way RM ANOVA, treatment effect: $F_{(1,12)} = 4.757$, $p = 0.0498$; interaction effect: $F_{(26,312)} = 4.120$, $p < 0.0001$; Sidak's *post-hoc*: $p < 0.05$ for 10-40 min; **Fig. 3H**; t -test, $t_{(23)} = 2.794$, $p = 0.0162$; **Fig. 3H** inset). However, short-term plasticity measured by PPF decreased during the initial 30 min of recording (Two-way RM ANOVA, interaction effect: $F_{(26,312)} = 3.457$, $p < 0.0001$; Sidak's *post-hoc*: $p < 0.05$ for 20-30 min; **Fig. 3I**), possibly reflecting a ceiling effect on fPSP amplitude after LTP induction. With these results, we demonstrate that, although ELS does not affect basal synaptic efficiency, it significantly increases the ability of HPC-PFC circuits to undergo LTP.

ELS induces long-term neural inflammation, but not neuronal loss

We sought to investigate histopathological alterations in cortico-limbic circuits that could underlie the abnormalities in behavior and HPC-PFC synaptic plasticity of ELS rats (**Fig. 4A**). In accordance with previous findings, ELS rats did not present neuronal loss when adults (Mixed-effects model, $F_{(1,12)} = 0.7683$, $p = 0.3980$; uncorrected Mann-Whitney test, $p = 0.1666$ -0.9372; **Fig. 4B**). Also, ELS did not produce long-term effects in the expression of PV (Mixed-effects model, $F_{(1,13)} = 0.008$, $p = 0.9297$; uncorrected Mann-Whitney test, $p = 0.3659$ -0.9307; **Fig. 4C**) or mGluR5 receptors (Mixed-effects model, $F_{(1,11)} = 0.0011$, $p = 0.9735$; uncorrected Mann-Whitney test, $p = 0.2467$ -0.8181; **Fig. 4D**). Finally, we found a strong increase in neuroinflammation, verified by the marked expression of GFAP in ELS rats in all regions investigated except the infralimbic cortex (Mixed-effects model, $F_{(1,18)} = 5.368$, $p = 0.0325$; uncorrected Mann-Whitney test, $p = 0.0067$ -0.0419 for GL, CA1, EC, PL, and TRN, and $p = 0.2298$ for IL; **Fig. 4E**). These results show that ELS produces subsequent brain inflammation. Also, important cognitive and behavioral impairments can manifest even in the absence of substantial neuronal loss in the HPC or PFC.

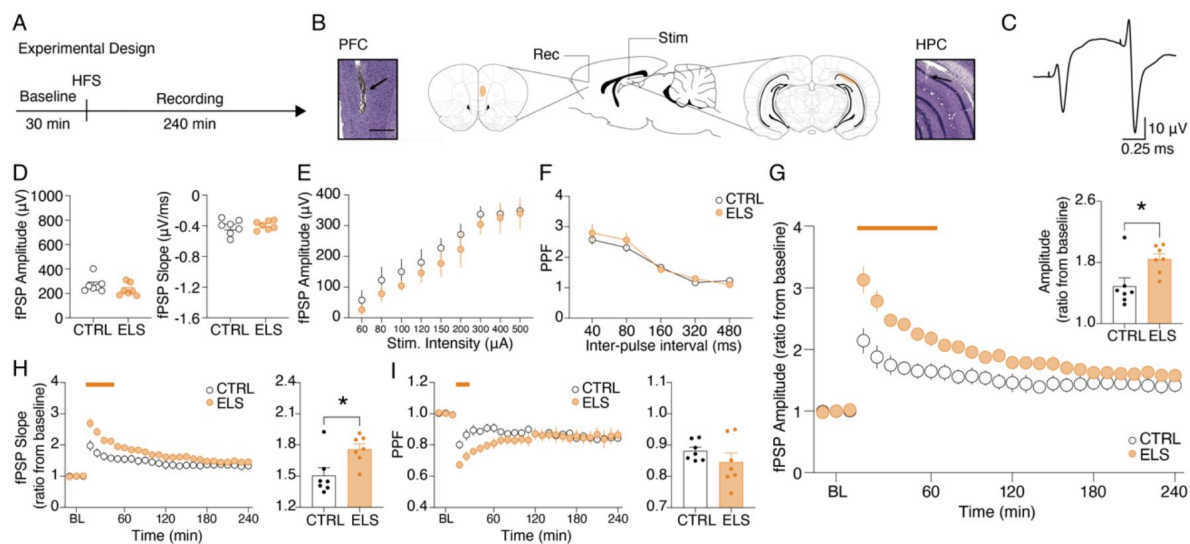


Figure 3.

Aberrant HPC-PFC synaptic plasticity in animals submitted to ELS.

A, Experimental design including high-frequency stimulation (HFS) to LTP induction. **B**, Electrode placement and representative electrolytic lesions in Cresyl Violet stained coronal sections. **C**, Representative evoked fPSPs in the PFC. No difference between ELS and CTRL rats in terms of basal neurotransmission and synaptic efficacy, as shown by: **(D)**, fPSP amplitude (left) and slope (right); **(E)**, input-output curve; and **(F)**, paired-pulse facilitation (PPF) curve. **G**, Despite no changes in basal synaptic efficacy, ELS rats showed aberrantly increased LTP, represented by higher fPSP amplitude ratios from baseline (BL) post-HFS, both across time blocks (main curve graph) and their averages (bar graph, inset). **H**, As in **G**, but from fPSP slopes. **I**, ELS rats showed reduced PPF in the initial 30 min (curve graph). Averaged data resulted in no differences between the groups (bar graph, inset). † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Error bars represent the mean $\pm$ SEM.

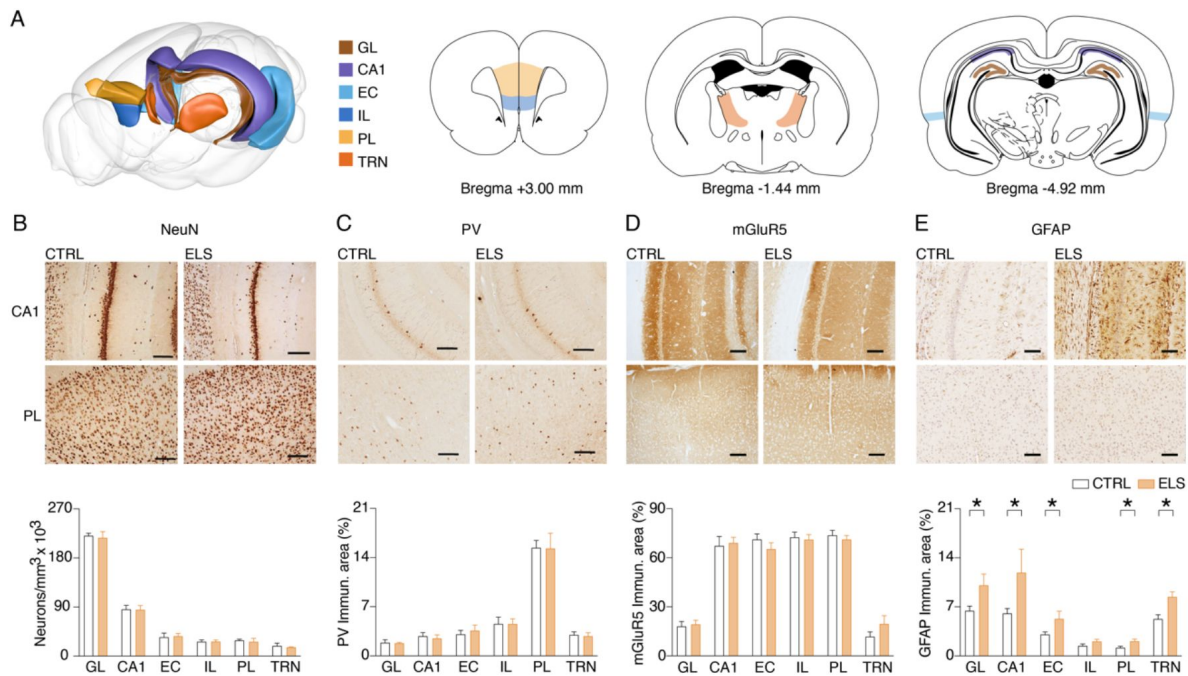


Figure 4.

ELS induces long-term neural inflammation, but not neuronal loss.

A 3-D brain diagram and coronal sections representing the brain regions investigated. **B-E**, Representative photomicrographs from immunohistochemistry each pair of columns represents a group comparison (left CTRL and right ELS) and each line is related to CA1 and PL, respectively. **B**, NeuN immunohistochemistry showing that ELS rats did not present neuronal loss. **C**, Parvalbumin (PV) immunohistochemistry. **D**, mGluR5; **E**, GFAP, showing increased astrogliosis induced by ELS. GL: Granule layer of dentate gyrus; CA1: *Cornu Ammonis* 1; EC: Entorhinal cortex; IL: Infralimbic cortex; PL: Prelimbic cortex; TRN: Thalamic Reticular Nucleus. * $p < 0.05$ Error bars represent the mean $\pm$ SEM.

Cognitive impairment and sensorimotor disturbances are distinctly correlated with abnormal LTP and neuroinflammation

By understanding the association between behavioral and neurobiological variables, we would be able to infer putative mechanisms underlying the behavioral comorbidities observed in ELS animals. To investigate multivariate relationships between the behavioral (cognitive and sensorimotor factors) and neurobiological (synaptic plasticity and immunohistochemical measures) dimensions of ELS effects, we conducted CCA (**Fig. 5A**). CCA produced a significant high-correlation model ($r = 0.94$, Hotelling-Lawley trace test, F -approximation_(10,12) = 3.2311, $p = 0.0164$; **Fig. 5B**) that clearly separated CTRL from ELS (behavior canonical dimension, t -test, $t_{(11)} = 3.693$, $p = 0.0035$; neurobiology canonical dimension, t -test, $t_{(11)} = 3.917$, $p = 0.0024$; **Fig. 5C**). By analyzing the coefficients, we observed that the behavior canonical dimension showed higher values for the sensorimotor factor, but also included the cognitive factor, although at a lower degree (**Fig. 5D**). The neurobiology canonical dimension showed a larger value for the GFAP, intermediate values for LTP and PV and the lowest coefficients for NeuN and mGluR5 (**Fig. 5D**). A similar pattern was observed when analyzing the loadings for each dimension (**Fig. 5E**). Notably, the neurobiological variables have only weak correlations between them (**Fig. 5F**). To further dissect specific multivariate relationships between behavioral and neurobiological measures, we constructed generalized linear models (GLM). The GLM indicates GFAP, PV, and LTP as the most significant variables that explain the sensorimotor factor variance (coefficients = 2.3775, 0.8457, and 0.5187; $p = 0.1164$, 0.00548, 0.08474 for GFAP, PV, and LTP, respectively, (**Fig. 5G**). Accordingly, GLM multivariate regression indicates a strong association between the sensorimotor factor and all significant GLM coefficients (GFAP, PV, plus LTP: $r^2 = 0.7974$, $p = 0.0116$; **Fig. 5H**), as well as sensorimotor factor and univariate neurobiological measures (GFAP: $r = 0.6840$, $p = 0.0099$; **Fig. 5I**, PV: $r = 0.6548$, $p = 0.0151$; **Fig. 5J**). Strikingly, we found a quadratic relationship between synaptic plasticity effects and measures of cognitive impairment. Poor performance on the working memory test was observed in rats that presented too low or too high values of HPC-PFC LTP, suggesting a U-shaped relationship between LTP and cognitive performance (**Fig. 5K-L**). We observed this quadratic relationship using both the cognitive factor (gaussian GLM, LTP coefficient = 1.22, $p = 0.0162$; **Fig. 5K**) and raw error values from the radial maze test using a quasi-Poisson GLM ($p = 0.00843$; **Fig. 5L**).

Taken together, both neuroinflammation and synaptic plasticity alterations may underlie the behavioral effects of ELS, with neuroinflammation being more associated with the sensorimotor dimension and LTP with the cognitive dimension.

ELS animals present epileptiform-like activity when adults

Given the results above, we decided to further investigate the relationships between circuit dynamics and behavioral dysfunctions during different brain states using chronic electrophysiology recordings. We classified LFP epochs into active wake (ACT), non-REM (NREM) sleep, rapid eye movement sleep (REM), and quiet wake (QWK; **Fig. 6A-D**). No CTRL vs. ELS differences were found in sleep architecture, according to the mean consecutive duration of each state, percentage of time awake, number of persistent episodes in each state (Two-way RM ANOVA, $F_{(1,13)} = 0.6500$, $p > 0.4346$; **Fig. 6E-G**), and percentage of time in each state (Two-way RM ANOVA, $F_{(1,13)} = 0.5425$, $p > 0.4745$; **Fig. 6H**).

We also examined if ELS animals presented ictal activity during recordings. We found that 44.45% of the ELS animals presented electrographic discharges similar to spike-wave discharges (SWD) or rhythmical spiking (**Fig. 6H-I**). SWDs occurred mainly during NREM sleep and less often in QWK and ACT but rarely occurred during REM epochs (**Fig. 6J-M**). The SWDs were not found in the CTRL rats, indicating that ELS induces an epileptogenic process.

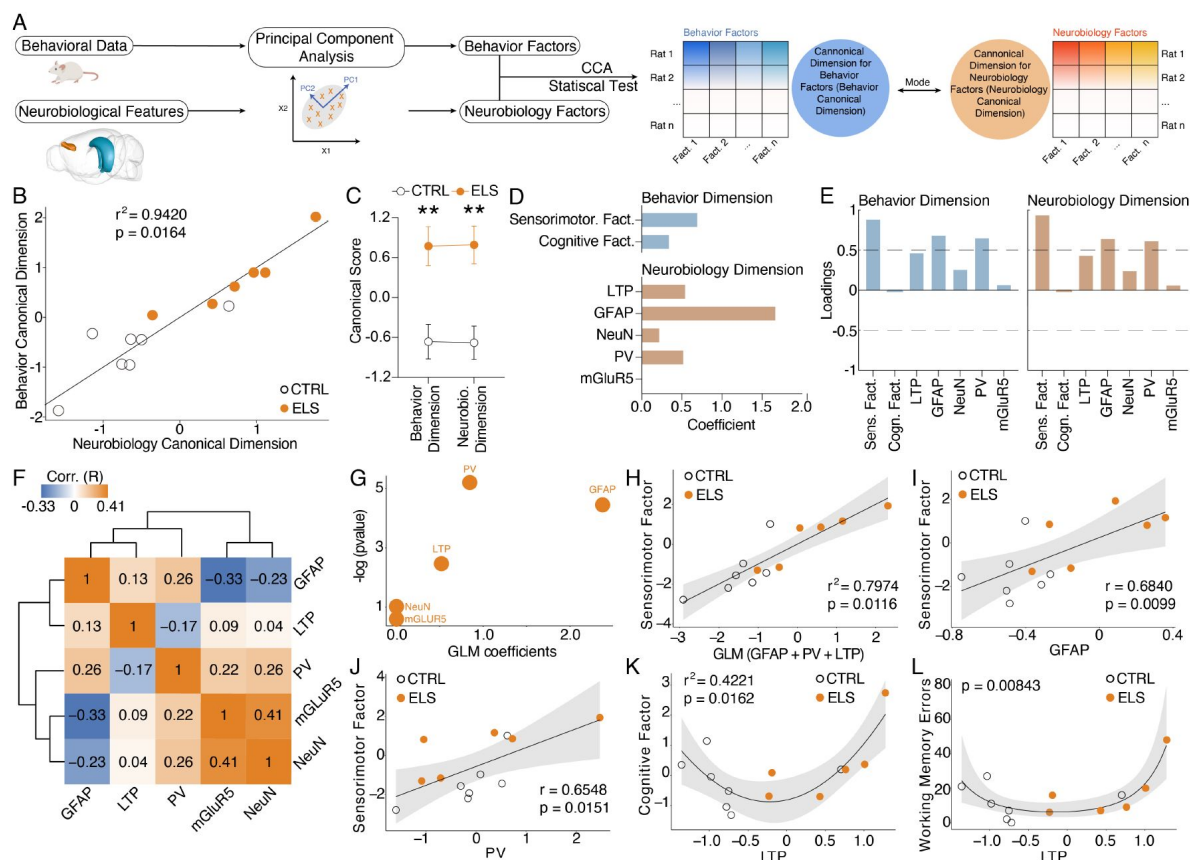


Figure 5.

Cognitive impairment and sensorimotor deficits are distinctly correlated with abnormal LTP and neuroinflammation.

A, Steps performed in data processing and analysis for canonical correlation analysis (CCA). **B**, CCA analysis showing CTRL and ELS animals projected onto the first canonical correlation. We found a strong correlation between the neurobiology canonical dimension and the behavior canonical dimension. **C**, Canonical scores showing the clear CTRL vs. ELS difference that the CCA was able to capture. **D**, Coefficients for behavior and neurobiology canonical dimensions, showing greater contributions of the sensorimotor factor, GFAP, PV, and LTP for the correlation. **E**, Loadings of the behavioral and neurobiological variables on each canonical correlation dimension indicating an association between GFAP and PV with the sensorimotor factor. **F**, Correlation matrix showing only weak correlations between the neurobiological features. **G**, GLM regression between neurobiological features and the sensorimotor factor. **H**, GLM model between neurobiological features and the sensorimotor factor show a strong association with GFAP, PV, and LTP. **I**, GLM regression between LTP and the cognitive factor shows a quadratic association (left). This quadratic association was further confirmed performing a quasi-Poisson regression between LTP and raw values of working memory errors (right). ** $p < 0.01$. Error bars represent the mean $\pm$ SEM.

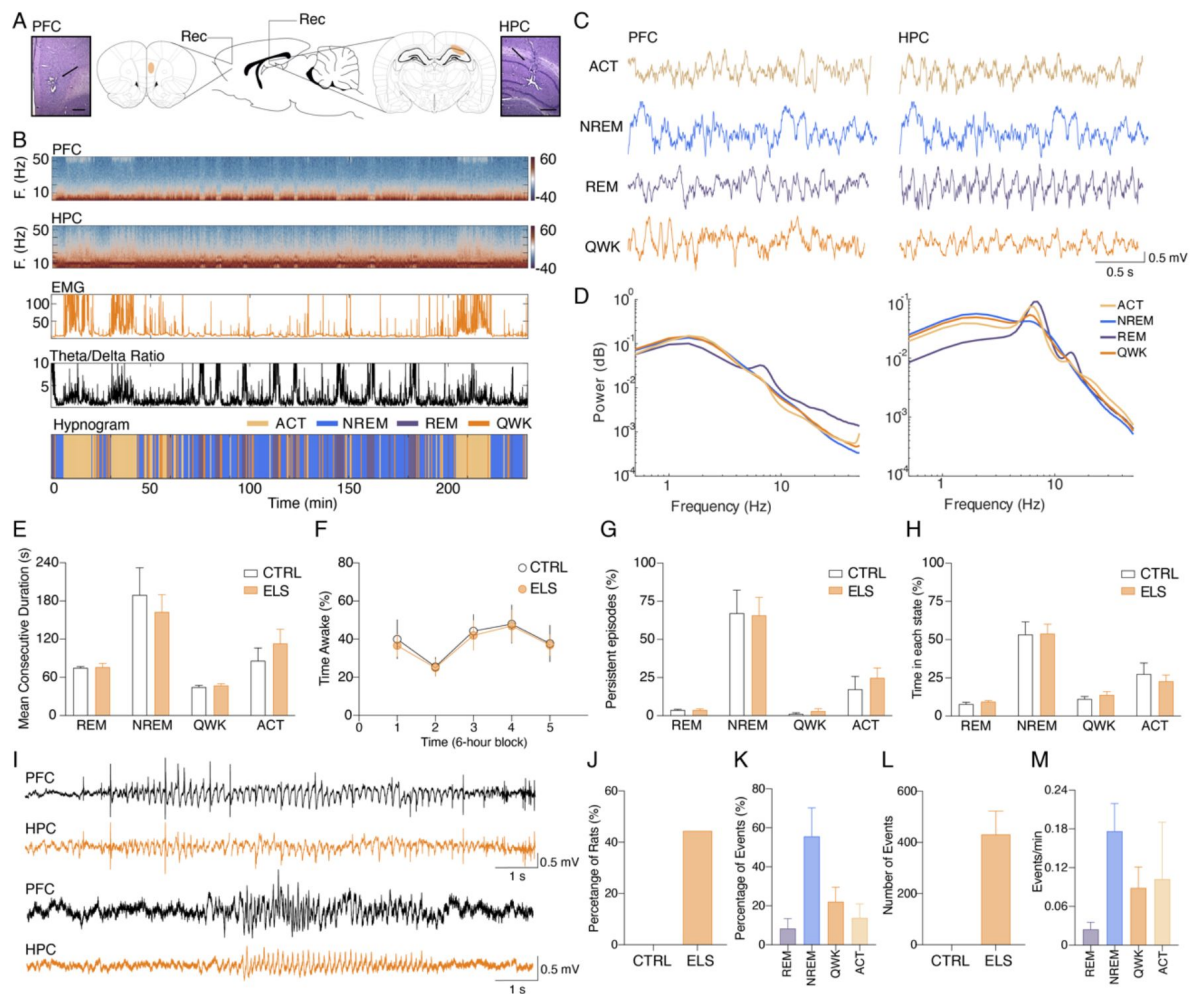


Figure 6.

Classification of sleep-wake states and electrographic discharges in ELS rats.

A, Electrode placement and representative electrolytic lesions in Nissl-stained coronal sections. **B**, Representative spectrogram, electromyogram (EMG) root mean squared (RMS), and HPC theta/delta ratio used to score the sleep-wake cycle and to generate the hypnogram (bottom). **C**, Representative LFPs (top) recorded from PFC and HPC during ACT, NREM, REM, and QWK. **D**, Average power spectral density for each state in the PFC and HPC of a representative CTRL animal. Note the increase in HPC theta power during ACT and REM. ELS rats did not present alterations in the sleep-wake cycle. **E-H**, There were no differences between ELS and CTRL in the **(E)** Mean duration of consecutive epochs for each state, **(F)** percentage of time awake across 6-h blocks, and in the **(G)** number of persistent episodes (episodes lasting longer than 3 s, top) or the **(H)** percentage of time spent in each state (bottom). **I**, Representative LFP from PFC and HPC showing a representative spike-wave discharge (SWD, top) and a poly-spike discharge (bottom). **J**, Electrographic discharges were found in 44.5% of ELS animals and were absent in CTRL animals. **K**, Percentage of events occurring in each sleep-wake stage. **L**, Average total number of SWD events in CTRL and ELS animals. **M**, Frequency of SWDs occurring per state. SWDs events rarely occurred during REM. Error bars represent the mean $\pm$ SEM.

Increased HPC theta power during behavioral activity in ELS animals

We initially concentrated our investigation on the ACT behavioral state to associate oscillatory patterns with the behavioral abnormalities observed. We found that ELS animals presented a higher theta power during ACT states in the HPC (t -test, $t_{(13)} = 2.259$, $p = 0.0417$; **Fig. 7A-D**), but not in the PFC (t -test, $t_{(13)} = 0.2636$, $p = 0.7961$; **Fig. 7B**). This increased theta activity is more prominent during the first periods of recording and exploration of the new environment (linear mixed-effects model, $t = 2.10$, $p = 0.055$; Uncorrected t -tests for period comparison: $t_{(13)} = 2.145$, 2.367 , and 2.319 , $p = 0.051$, 0.034 , and 0.037 for periods 1, 2, and 3, respectively; **Fig. 7E**). To further explore whether theta power increase reflects a true oscillation and is not due to alterations in the aperiodic components of the PSD, we also performed a PSD parametrization to separate the theta oscillation (**Fig. 7F**). We found an increase in the theta oscillatory activity in ELS animals during ACT states, indicating an actual oscillatory effect (Mann-Whitney, $U = 7$, $n = (6,9)$, $p = 0.0176$; **Fig. 7G-H**). Hippocampal theta is well known to occur during periods of locomotor activity, e.g., postural changes, head movements, and rearing (Buzsáki and Moser, 2013) representing a possible confounder. To control for this, we co-examined HPC theta power and EMG distributions during active states in CTRL vs. ELS rats (**Fig. 7I**). Indeed, theta oscillatory increase in ELS animals did not appear to be related to more movement, as we did not observe a significant association between EMG and theta power. In contrast, we found higher theta power in ELS animals even at low levels of EMG activity (**Fig. 7I**).

ELS promotes abnormal gamma coordination in the PFC during behavioral activity

Considering the extensively documented involvement of PFC gamma activity in cognitive and sensory processes, as well as its alterations observed in animal models of neurodevelopmental and psychiatric disorders such as ASD and schizophrenia (Rojas and Wilson, 2014; Sigurdsson, 2016), we investigated the possible effects of ELS on PFC gamma activity during active states. ELS animals showed a reduced theta-gamma amplitude correlation during ACT states (t -test, $t_{(13)} = 2.851$, $p = 0.0136$; **Fig. 8A-C**), indicating that high-gamma activity is less coordinated, despite the increase in theta power. Interestingly, there were no increases in both low or high-gamma power (low-gamma: Mann-Whitney, $U = 16$, $n = (6,9)$, $p = 0.2238$; high-gamma: Mann-Whitney, $U = 24$, $n = (6,9)$, $p = 0.7756$; **Fig. 8D**). Amplitude correlation is an indicator of large-scale cortical interactions that govern cognition (Siegel et al., 2012). Thus, the lower theta-high gamma correlations found here may be related to the cognitive disruptions caused by ELS.

Next, we found a reduction in the PFC spectral exponent in ELS animals assessed by aperiodic PFC activity during ACT state (note the flatter spectra of ELS in **Fig. 8E**), distribution of the aperiodic component (exponent and offset, **Fig. 8F**), and comparison in exponent parameter between groups (t -test, $t_{(13)} = 2.010$, $p = 0.0656$; **Fig. 8G**). Reductions of the spectral exponent have been discussed to indicate increased excitation/inhibition imbalance (Gao et al., 2017). This abnormal gamma dynamics in the ELS indicates a less coordinated oscillatory activity, which may reflect impaired limbic-cortical or cortico-cortical interactions that underlie cognitive dysfunctions.

ELS impairs HPC-PFC coherence during REM sleep

HPC-PFC connectivity and information flow during REM sleep is essential for memory consolidation (Lesting et al., 2013). Because we found cognitive impairments in ELS animals, we investigated HPC-PFC interactions during REM. First, we observed that the theta power in the PFC and HPC during REM is not different between groups (PFC: t -test, $t_{(13)} = 1.320$, $p = 0.2096$; HPC: t -test, $t_{(13)} = 0.3557$, $p = 0.7278$; **Fig. 9A-B**). However, we found that ELS rats show reduced HPC-PFC connectivity in the theta band, measured by spectral coherence during REM sleep (t -test, $t_{(13)} = 2.183$, $p = 0.0480$; **Fig. 9C-D**). This difference is present throughout the entire recording, as

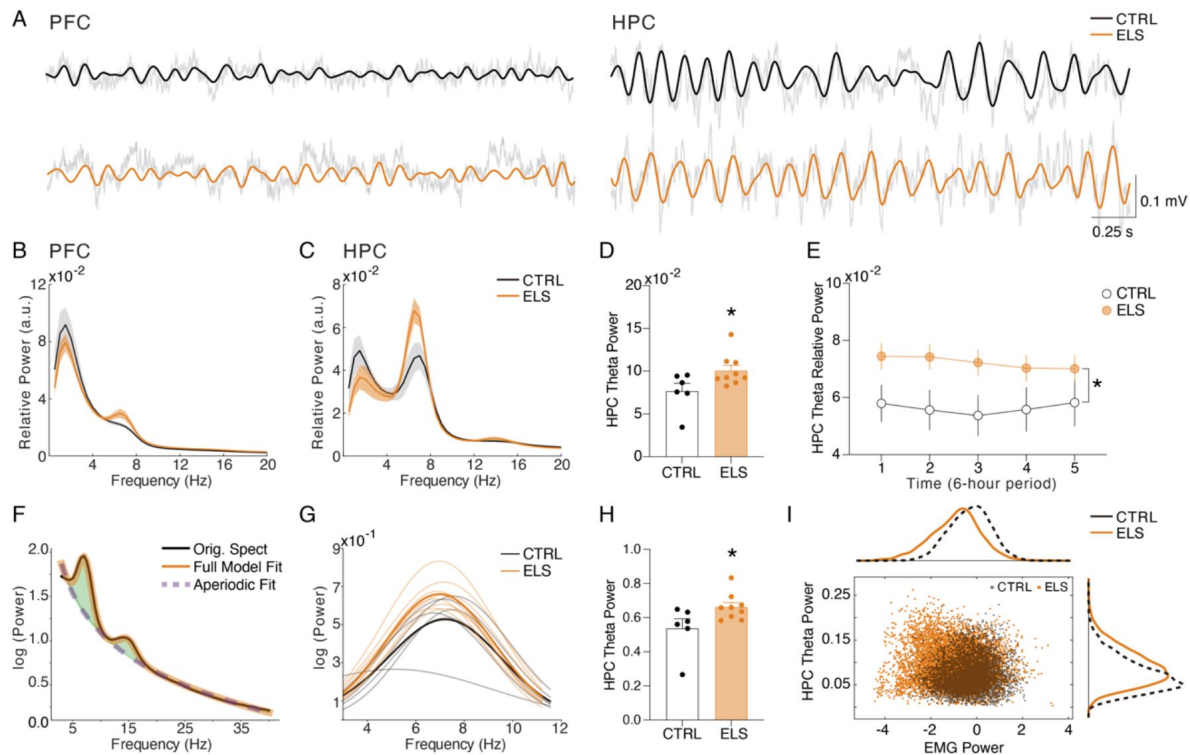


Figure 7.

Increased HPC theta power during behavioral activity in ELS animals.

A, Representative raw LFPs (transparent black lines) and theta-filtered signal of CTRL (black) and ELS (orange) from the PFC (left) and HPC (right) during ACT. **B,** Relative PSD of the PFC during ACT. **C,** Relative PSD of the HPC during ACT. **D,** ELS rats present an enhanced HPC theta power during ACT. **E,** Theta relative power during each of the 6-hour periods of recording, showing a general increase in the ELS animals that is more pronounced in the initial 18 h of recording. **F,** Parametric model used to distinguish oscillatory from aperiodic component. **G,** Oscillatory components extracted from ACT PSDs used to calculate theta power. **H,** Theta power increase in ELS rats is due to a true oscillatory component and not from the aperiodic activity in ACT states. **I,** Theta power increase in ELS rats is independent of locomotor activity. Note from the distribution of relative theta power and Z-scored EMG (trapezius) values for all rats that EMG power is not related to the regions of greater theta power in the ELS animals. * $p < 0.05$. Error bars represent the mean $\pm$ SEM.

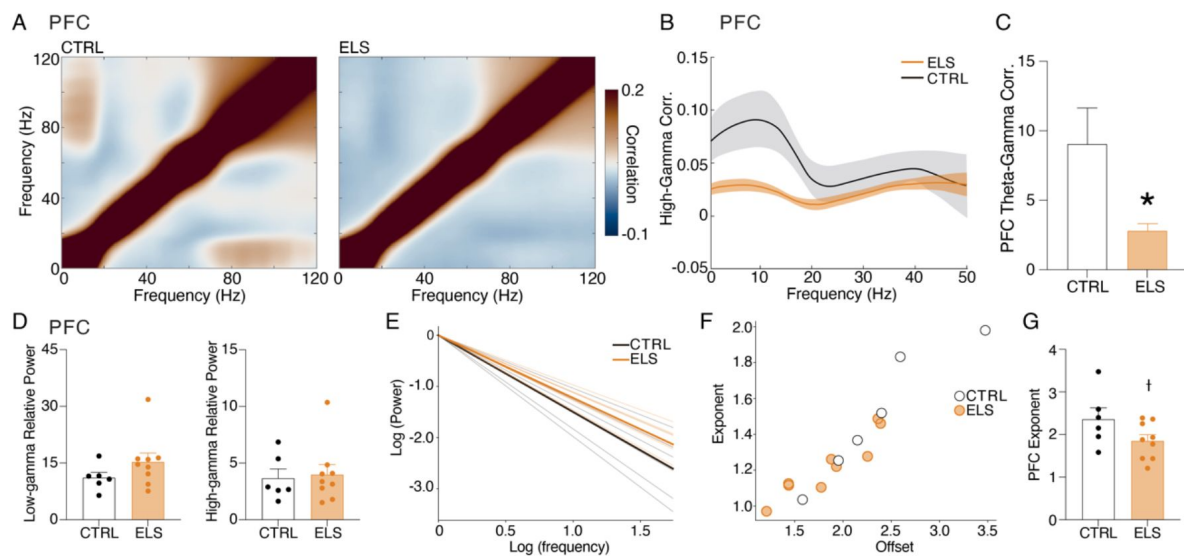


Figure 8.

ELS promotes abnormal gamma coordination in the PFC during behavioral activity.

ELS rats present an abnormal gamma activity in the PFC. **A**, Grand-average amplitude correlogram for CTRL (left) and ELS (right) group. Note the absence of amplitude correlation between theta and gamma activity in the ELS group. **B**, High-gamma (65-100 Hz) correlation with low-frequency activity. Note the peak around theta oscillation in the CTRL group and the absence in ELS. **C**, ELS present an impaired theta-gamma amplitude correlation. **D**, There is no difference between the low-(top) and high-gamma (bottom) average power. **E**, Aperiodic activity extracted from the PFC of each rat during ACT state. **F**, Distribution of the parameters of the aperiodic component (exponent and offset) for each rat. **G**, ELS rats show a tendency to reduction ($p = 0.0656$) in the exponent parameter of the aperiodic component, as illustrated by the flatter spectra. † $p < 0.1$, * $p < 0.05$. Error bars represent the mean $\pm$ SEM.

observed by the comparisons for all 6-hour periods (linear mixed-effects model, $t = -2.27$, $p = 0.0408$; **Fig. 9D** [↗](#), right). We further confirmed this difference by analyzing the cross-correlation and cross-spectrum power of the HPC-PFC LFP during REM (**Fig. 9E-G** [↗](#)). Cross-correlation analysis reveals a clear directionality between signals from the HPC to the PFC in both groups that happens markedly at a specific lag (-80 to -20 ms; **Fig. 9E** [↗](#)), in which ELS rats present the most profound decrease in the correlation peak (t -test, $t_{(13)} = 2.533$, $p = 0.0250$; **Fig. 9F-G** [↗](#)).

We noticed from inspecting the theta coherence changes between ACT and REM – the two states with the strongest theta coherence – that all CTRL animals exhibited greater theta coherence during REM (paired t -test, $t_{(5)} = 8.023$, $p = 0.0005$; **Fig. 9H** [↗](#)), while in the ELS group, part of the animals did not present such an increase (paired t -test, $t_{(8)} = 1.924$, $p = 0.09$; **Fig. 9H** [↗](#)). This dynamic is further illustrated by observing the coherence changes across REM, QWK, and ACT states. While most CTRL rats present an inclined decrease from REM to QWK and a slight increase from QWK to ACT, ELS rats display a flatter curve (**Fig. 9I** [↗](#)). By quantifying the homogeneity of the distributions of average theta coherence values across states with entropy, we found a significantly higher entropy (more homogeneous) in ELS than CTRL (paired t -test, $t_{(13)} = 2.695$, $p = 0.0184$; **Fig. 9J** [↗](#)). These results indicate that ELS promotes impairment in HPC-PFC connectivity during REM sleep and in the regulation of state-dependent dynamics.

ELS rats display REM-like oscillatory dynamics during active behavior

After we noticed that ELS animals show less distinction of some electrophysiological activities between states, particularly REM and ACT, we decided to further characterize the long-term effects of ELS on HPC-PFC state-dependent oscillatory dynamics across the sleep-wake cycle. For that, we constructed a state map based on two spectral power ratios (Dzirasa et al., 2006 [↗](#); Gervasoni et al., 2004 [↗](#)). This state map framework allows consistent characterization of all major brain states, revealing a dynamic global structure produced by the collective activity of forebrain structures (Gervasoni et al., 2004 [↗](#)). First, we verified that the state maps reflected other brain and muscle activities that were not directly used for the map construction, such as the concentrated distribution of the PFC delta power during NREM, of the HPC theta/delta ratios in REM, of the EMG power in awake states, and of HPC-PFC theta coherence in both REM and awake states (**Fig. 10A** [↗](#)). Thus, as expected, our state maps presented three distinct clusters that precisely located REM, NREM, and ACT epochs (**Fig. 10B** [↗](#)).

Using this approach, a previous study reported an overlap between awake and REM states in a hyperdopaminergic mouse model (Dzirasa et al., 2006 [↗](#)). Because we found ELS-related functional changes specifically during ACT and REM, we sought to investigate the convergence between these two states. Remarkably, we also found that ELS rats showed a significant similarity between the REM and ACT states (**Fig. 10B** [↗](#)). While in CTRL rats the distributions of Euclidean distances were larger, homogeneous, and symmetrical (**Fig. 10C** [↗](#)), ELS rats presented highly variable patterns across animals, most of which with skewed distributions concentrated at shorter distances (**Fig. 10C** [↗](#)). Indeed, the Euclidean distance was significantly reduced in ELS (t -test, $t_{(13)} = 2.326$, $p = 0.0369$; **Fig. 10D** [↗](#)), suggesting that the spectral features of REM and ACT are more similar in ELS animals.

Interestingly, we observed that this difference in states similarity was stronger in the initial 12 hours of recording, which corresponds to the periods of habituation to the novel environment (t -test, for periods 1-2: $t_{(13)} = 2.45$, $p = 0.029$; for periods 3-4: $t_{(13)} = 0.9409$, $p = 0.3638$; **Fig. 10E** [↗](#)).

To further demonstrate the higher similarity between REM and ACT states in ELS animals, we used machine learning algorithms in an attempt to separate these brain states through a *brute-force* approach. We observed that the two algorithms tested had a worse discrimination performance in the ELS rats, as verified by the smaller AUC-ROC (t -test, for SVM: $t_{(198)} = 99.59$, $p < 0.0001$; for

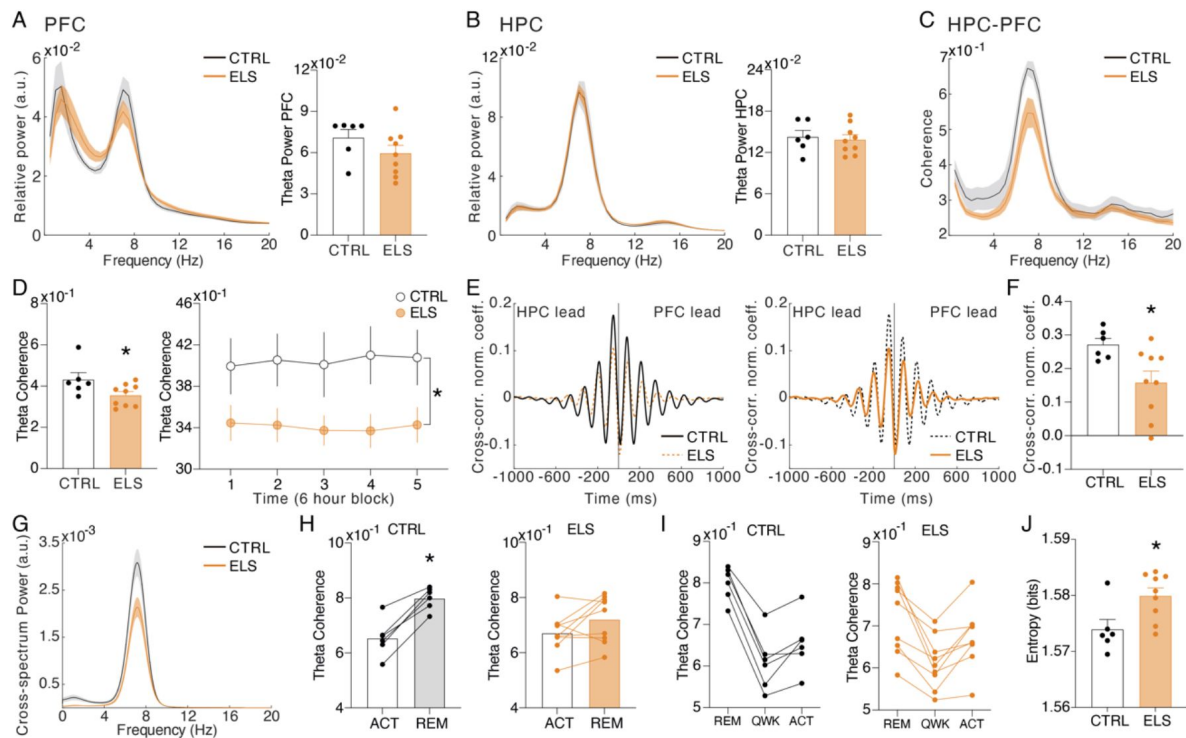


Figure 9.

ELS impairs HPC-PFC coherence during REM sleep.

A, PSD in the PFC during REM. **B**, PSD in the HPC during REM. No significant alterations in theta power during REM are observed in ELS rats. **C**, Theta coherence is diminished in ELS rats. **D**, Significant effect on the HPC-PFC theta coherence by averaging the entire recording (left) and across all periods (right). **E**, Time domain cross-correlation between PFC and HPC LFPs during REM epochs highlighting CTRL (left) and ELS (right) group averages. **F**, Reduction in the peak of correlation between -80 to -20 ms in ELS rats. **G**, Average cross-spectrum power density showing a reduced theta peak in the ELS rats. **H**, Theta HPC-PFC coherence increases from ACT to REM states. We can observe that all the CTRL rats present an increase from ACT to REM, while in ELS rats this is not observed. Paired *t*-test $*p < 0.05$. **I**, Theta coherence dynamics across the sleep-wake cycle. CTRL rats display a homogeneous dynamics with theta coherence being maximal at REM, minimal in QWK, and intermediate in ACT states. This dynamic is disrupted in ELS rats that display a higher variance between animals, with some animals displaying similar coherence in REM and ACT. **J**, Entropy of the theta coherence dynamics is higher in ELS rats, representing the more homogeneous values across states.

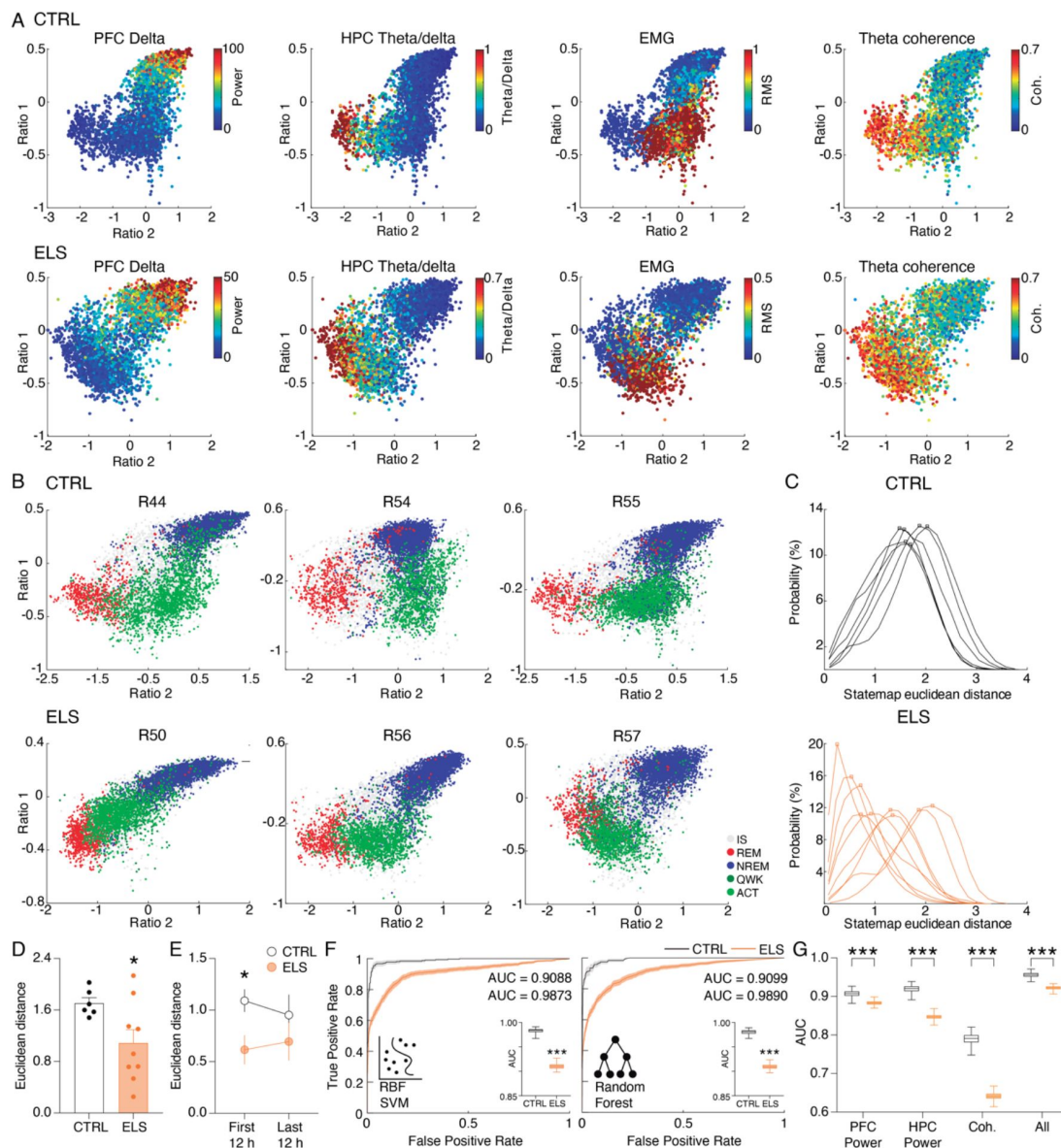


Figure 10.

ELS rats display REM-like oscillatory dynamics during active behavior.

A, State map construction reflects the sleep-wake cycle and spectral dynamics for one representative CTRL (top) and ELS (bottom) rat. PFC delta power is evident during NREM, theta/delta ratio is maximum in REM, EMG power describes ACT states, and HPC-PFC theta coherence is stronger close to REM. Note that in the ELS rat, the coherence is also broadly high during the ACT state. **B**, Representative state maps from three CTRL (top) and three ELS (bottom) rats. CTRL rats displayed a clear separation between REM (red), ACT (light green), and NREM (blue) clusters. ELS displayed an overlap between REM and ACT states. **C**, Distributions of the Euclidean distances between each REM and ACT epoch for CTRL (top) and ELS (bottom) rats. ELS distributions were heterogeneous, many of which were skewed to shorter distances. **D**, Average Euclidean distance showing that ELS rats have REM-like oscillatory dynamics during active behavior. **E**, REM-like oscillations during ACT were more pronounced in ELS during the initial 12 h of recording. **F**, Machine learning approach to decode REM *versus* ACT states. We used radial basis function support vector machine (SVM) and random forest (RF) algorithm in order to classify REM and ACT epochs in CTRL (black) or ELS (orange) rats. Using a bootstrapped confidence interval, we observed that the discriminative performance was worse in ELS animals, even when using “brute-force” algorithms. **G**, A machine learning algorithm (random forest) fitted on whole-spectrum data also shows a significantly worse performance in ELS for PFC power, HPC power, coherence, and all estimates combined. † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Error bars represent the mean $\pm$ SEM.

random forest: $t_{(198)} = 107.6$, $p < 0.0001$; **Fig. 10F** [↗](#)). This worse discriminative performance also arises when using random forest algorithm (which presents the best discriminative performance) on the original whole-spectrum high-dimensional variables (t -test, for PFC power: $t_{(198)} = 21.34$, $p < 0.0001$; for HPC power: $t_{(198)} = 58.78$, $p < 0.0001$; for coherence: $t_{(198)} = 80.82$, $p < 0.0001$; for all the spectral estimates combined: $t_{(198)} = 39.23$, $p < 0.0001$; **Fig. 10G** [↗](#)). This lack of discriminative performance once again corroborates spectral similarities between REM and ACT states in ELS rats.

ELS rats present behavioral and neurochemical sensitivity to dopaminergic activation

REM-like neural oscillations during the awake state were described in pharmacological and genetic animal models of hyperdopaminergia (Dzirasa et al., 2006 [↗](#)). In this context, we conducted a final experiment to investigate the susceptibility of ELS rats to dopaminergic activation using apomorphine (unspecific dopaminergic agonist). We examined behavioral sensitivity by comparing the locomotion produced by apomorphine in the open field (**Fig. 11A** [↗](#)). Our data demonstrate that apomorphine produces a higher increase in locomotion in ELS rats when compared to CTRL across time blocks (Two-way RM ANOVA, treatment effect: $F_{(1,27)} = 4.260$, $p = 0.0487$, Sidak's *post-hoc*: 5 min $p = 0.0376$; **Fig. 11B** [↗](#)) or between average normalized distances (Mann-Whitney, $n(CTRL, ELS) = 14/15$, $U = 32$, $p = 0.0009$; **Fig. 11B** [↗](#) inset).

Immediately after the behavioral test, we collected samples of NAc, PFC, and HPC to quantify neurotransmitters. The NAc was included in this analysis because it is highly implicated in dopamine sensitivity and exploratory behavior (Salomone et al., 2012). Our data demonstrate that apomorphine reduces dopamine in the NAc of ELS rats, with no effects on dopamine turnover, as measured by DOPAC/DA ratio (Sidak's multiple correction procedure, $t_{(12)} = 3.656$, $p = 0.0033$; **Fig. 11C** [↗](#)). Still, a GLM model implicated NAc dopamine turnover as a significant predictor of apomorphine-induced hyperlocomotion (*coefficient* = -0.7108 , $p = 0.00297$; **Fig. 11D** [↗](#)). Consistently, apomorphine is known to reduce endogenous DA release and DA metabolism in rodents and humans (De La Fuente-Fernández et al., 2001 [↗](#); Ozaki et al., 1989 [↗](#)). We also observed a reduction in dopamine turnover in the HPC of ELS rats (Sidak's multiple correction procedure, $U = 9$, $p = 0.0164$; **Fig. 11E** [↗](#)) but no significant effects in the PFC.

Interestingly, we found that the HPC of ELS rats shows a higher Glu content and higher Glu/GABA ratio than CTRL animals following apomorphine injection (Sidak's multiple correction procedure, Glu: $t_{(15)} = 2.566$, $p = 0.0215$; Glu/GABA: $U = 9$, $p = 0.0079$; **Fig. 12F** [↗](#)) indicating possible disturbances in the excitation/inhibition balance.

HPC-PFC oscillatory dynamics discriminate ELS individuals

After identifying the relevance of specific electrophysiological variables throughout the study, we sought to investigate if collective patterns of the electrophysiological covariates across individuals could assemble a multidimensional oscillatory signature that distinguishes the ELS subjects. Based on relevance, particularity, and state specificity, for this investigation, we chose: (1) HPC theta power (**Fig. 7H** [↗](#)), (2) theta coherence entropy (**Fig. 9J** [↗](#)), (3) PFC exponent (**Fig. 8G** [↗](#)), (4) PFC theta-gamma correlation (**Fig. 8C** [↗](#)), (5) theta coherence (**Fig. 9D** [↗](#)), and (6) state map distance (**Fig. 10D** [↗](#)). By performing PCA, we observed that the first two principal components completely discriminated ELS from CTRL animals (**Fig. 12A** [↗](#)). This discrimination was further confirmed by fitting a linear discriminant classifier (traces on **Fig. 12A** [↗](#)) that achieved a 100% accuracy using the entire dataset and 80% accuracy on leave-one-out cross-validation.

Furthermore, unsupervised clustering almost completely discriminated ELS from CTRL rats dividing all the animals into two major clusters: a predominantly ELS cluster (81.82% of ELS animals) and a complete CTRL cluster (100% of CTRL animals; **Fig. 12B** [↗](#)). The hierarchical

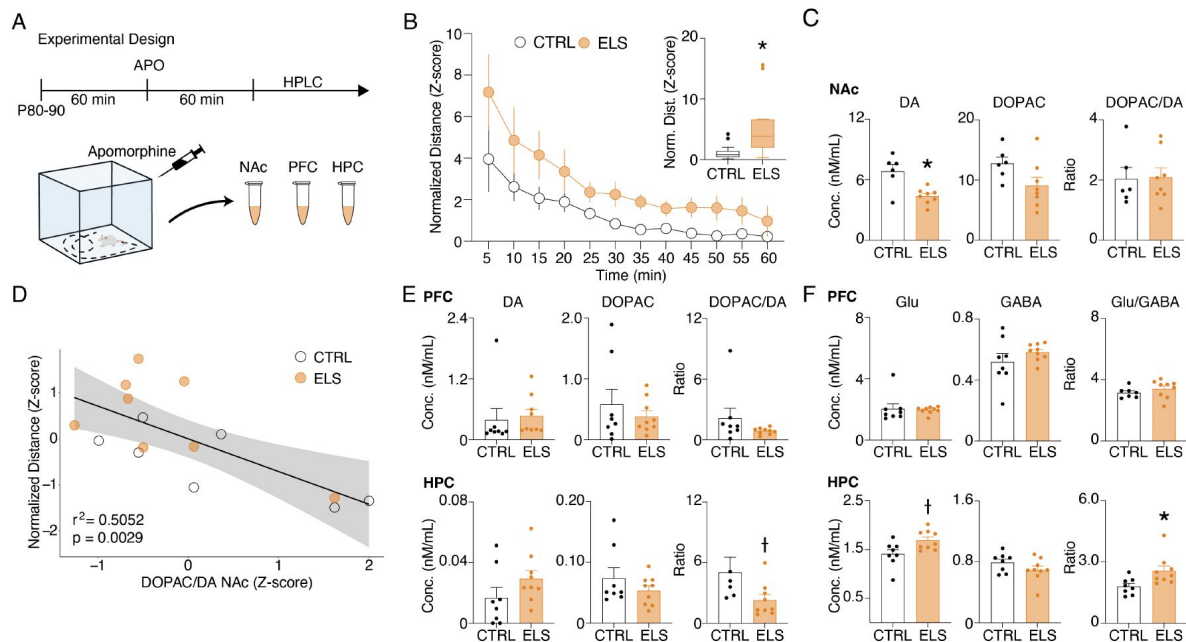


Figure 11.

ELS rats present behavioral and neurochemical sensitivity to dopaminergic activation.

A, Experimental design representing quantification of locomotion in the open field, apomorphine (APO) injection, and neurochemical analysis. **B**, Normalized locomotion across time blocks after APO injection, demonstrating increased locomotion of ELS rats induced by APO. Average normalized locomotion after APO injection (inset). **C**, Concentration of dopamine and metabolites in the NAc. **D**, GLM for the total distance after APO indicating DOPAC/DA ratio in the NAc as a strong significant variable associated with behavioral sensitivity to APO. **E**, Concentration of dopamine and metabolites (left) in the PFC (upper panel) and HPC (lower panel). Concentration of glutamate and GABA (right) in the PFC (upper panel) and HPC (lower panel). †Uncorrected $p < 0.05$, * multiple comparison corrected $p < 0.05$. Error bars represent the mean $\pm$ SEM.

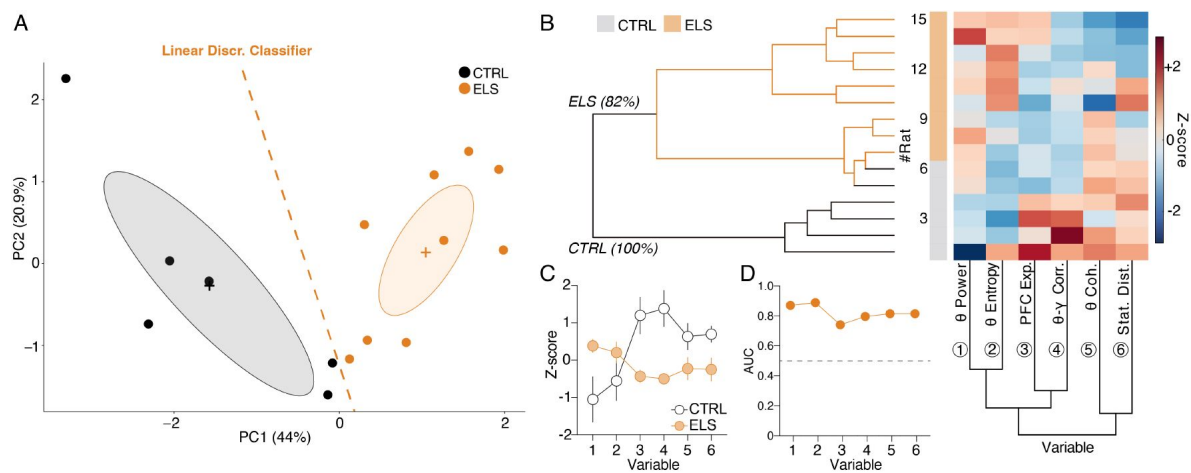


Figure 12.

HPC and PFC oscillatory dynamics discriminate ELS rats.

A, A linear discriminant classifier (dotted orange line) using the first two principal components (PCs) of relevant electrophysiological variables completely distinguishes ELS from CTRL rats (100% accuracy; 93.34% leave-one-out cross-validation). Note that the PC1 alone completely discriminates ELS rats. **B**, Unsupervised hierarchical clustering using the electrophysiological variables (bottom dendrogram) almost completely dissociates ELS from CTRL rats (left dendrogram). Inside the ELS cluster, we can observe one complete ELS branch, but the other overlapping with some CTRL animals, suggesting a graduality in the pathophysiological effects of ELS. **C**, Z-score values of each variable, showing distinct patterns of activity for each group. **D**, All the electrophysiological variables are good discriminators of CTRL and ELS rats (AUC-ROC > 0.7). † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Error bars represent the mean $\pm$ SEM.

clustering approach evidenced a progression of ELS-related oscillatory alterations, with a subcluster presenting an overlap between ELS and CTRL but the other with a more pronounced abnormal spectral fingerprint (**Fig. 12B** [↗](#)). In addition, the values of each variable are distinct in each group, which demonstrates that the electrophysiological variables used here are jointly affected by ELS at the functional level. Finally, we show that all the single electrophysiological variables have a high discriminative value ($AUC > 0.7$) (**Fig. 12D** [↗](#)), yet, their multivariate linear combination represented by PC1 scores (**Fig. 12A** [↗](#)) has the best classification performance with an AUC of 1.

Discussion

Here we showed that although ELS does not induce significant neuronal death, it leads to long-term cognitive impairments and behavioral abnormalities. These effects are correlated with neuronal inflammation, hypersensitive dopaminergic neurotransmission, and aberrant HPC-PFC synaptic plasticity. We also found that ELS rats exhibit brain state-dependent alterations – particularly during active behavior and REM sleep – characterized by increased theta power and impaired coordination of local and long-range activity. In addition, ELS presents an abnormal oscillatory state during active behavior oddly similar to REM sleep. Finally, the collective pattern of oscillatory alterations accurately distinguishes ELS from control animals.

Common pathophysiological mechanisms underlying ELS consequences and psychiatric-like impairments

Remarkably, the behavioral alterations found here are well-established translational features of neurodevelopmental disorders and psychiatric disorders related to epilepsy. Studies usually describe increased locomotion (Kubová and Mareš, 2013 [↗](#)) and working memory deficits (Tsai et al., 2012 [↗](#)) as consequences of ELS. In addition, we also found a reduced PPI, which is less studied but represents a well-established correlate of positive symptoms in clinical psychosis and sensorimotor alterations of ASD (Cheng et al., 2018 [↗](#); Turetsky et al., 2007 [↗](#); Ziermans et al., 2011 [↗](#)). Here, we presented a multivariate *sensorimotor factor* comprising a collection of behaviors frequently reported in animal models of schizophrenia, ADHD, and ASD (Ey et al., 2011 [↗](#); Regan et al., 2022 [↗](#); Sigurdsson, 2016 [↗](#)). Furthermore, we presented a *cognitive factor* mainly representing working memory deficits, a hallmark in several neurodevelopmental models and psychiatric comorbidities, particularly schizophrenia (Cope et al., 2016 [↗](#)). Importantly, the two factors were uncorrelated, but both resulted from ELS. This lack of association between cognitive and sensorimotor impairments suggests two neurobiological substrates, which are confirmed by our multivariate analysis, indicating the association of LTP to cognitive deficits and neuroinflammation to sensorimotor alterations.

Surprisingly, we found that cognitive impairment was associated with an abnormally increased susceptibility of HPC-PFC circuits to undergo LTP. This appears to be in contrast with previous studies, which reported reduced hippocampal LTP and cognitive deficits after ELS (Chang et al., 2003 [↗](#); O'Leary et al., 2016 [↗](#)). However, no hyperexcitability or spontaneous seizures were described in these studies (O'Leary et al., 2016 [↗](#)). In turn, Notenboom et al. (2010) [↗](#) used an animal model of febrile ELS and demonstrated increased LTP in HPC slices related to the reduction of seizure threshold, hyperexcitability, and spontaneous seizures later in life (i.e., epileptogenesis) (Dubé et al., 2006a [↗](#); Dube et al., 2000 [↗](#)). Indeed, paroxysmal activity has been shown to acutely increase synaptic efficacy and hippocampal LTP, thereby reducing the threshold for seizure induction (Ben-Ari & Gho, 1988 [↗](#); Minamiura et al., 1996 [↗](#)). In this context, the enhanced LTP observed in our data indicates a relationship with epileptogenesis, which concurs with the epileptiform discharges observed in ELS rats and the increases observed in the hippocampal Glu/GABA ratio.

We describe an inverted-U relationship between LTP and working memory, suggesting an optimal level of LTP susceptibility is required for adequate working memory performance. Aberrant synaptic plasticity could be a common mechanism between cognitive deficits and epileptogenesis induced by ELS. Hyperexcitable HPC-PFC circuits could facilitate the spread of epileptiform activity and promote alterations in synaptic saturation and excitatory/inhibitory balance, ultimately leading to cognitive deficits. In this sense, the abnormally increased LTP could impair sensory processes during working memory and explain the poor performance we observed in ELS rats. In a neural network with aberrantly enhanced LTP, synaptic plasticity could be biased toward potentiation, resulting in a lower ability to retain information or adapt to the environment (Navakkode et al., 2022 [DOI](#)). This rationale is supported by several genetic studies, including on risk genes for neuropsychiatric disorders and autism that described relationships between enhanced LTP and impaired learning (Garcia-Alvarez et al., 2015 [DOI](#); Meng et al., 2002 [DOI](#); Nuytens et al., 2013 [DOI](#); Yokoi et al., 2022 [DOI](#)). Interestingly, hyperdopaminergic signaling enhanced HPC-PFC LTP dose-dependently following an “inverted-U” function, where too high or too low dopamine levels disrupted LTP induction, indicating that altered DA neurotransmission could underlie the LTP abnormalities in ELS rats (Goto and Grace, 2006 [DOI](#); Otani et al., 2015 [DOI](#)).

We also report an increase in GFAP expression in ELS animals. Inflammatory processes initiated by a brain insult could lead to functional impairments and, ultimately, psychiatric symptoms (Vezzani et al., 2013 [DOI](#)). In ELS models, long-term glial activation has been related to behavioral impairments and epileptogenesis (De Oliveira et al., 2008b [DOI](#)), while treatment with inflammatory inhibitors resulted in reductions of GFAP and cognitive impairments (Abraham et al., 2012 [DOI](#); Somera-Molina et al., 2007 [DOI](#)). High levels of pro-inflammatory substances and neuroinflammation have been documented in animal models and patients with schizophrenia and ASD (Prata et al., 2017 [DOI](#); Vezzani et al., 2013 [DOI](#)). Noteworthy, the HPC of patients with temporal lobe epilepsy (TLE) and interictal psychosis present greater GFAP and microglia activation than TLE patients without psychosis (Kandratavicius et al., 2015). Additionally, GFAP is elevated in the frontal cortex of autistic subjects (Laurence and Fatemi, 2005 [DOI](#)). These findings are in accordance with clinical and preclinical biomarkers bridging psychosis and ASD to dopaminergic disturbances resulting from ELS.

In adult SE studies, behavioral abnormalities and cognitive deficits have been positively correlated to neuronal loss and spontaneous seizure frequency (Wolf et al., 2016 [DOI](#)). It is suggested that the etiological mechanisms of neuropsychiatric-related alterations and epileptic excitability spread progressively over time. Indeed, the behavioral impairments in ELS models tend to be more subtle than in animal models of TLE or hyperdopaminergia. It is possible that the observed alterations in neuroinflammation, neuronal dynamics, and neurochemistry represent initial epileptogenic pathological mechanisms that could progress to spontaneous seizures, progressive neuronal loss, and more intense behavioral manifestations (Dubé et al., 2006b [DOI](#); Kubová and Mareš, 2013 [DOI](#)).

Dysfunctional patterns of HPC-PFC oscillatory dynamics

Our results demonstrate a locomotion-unrelated increase in HPC theta power during active behavior in ELS animals. Theta oscillations underlie various cognitive processes and are believed to serve as a temporal organizer for multiple functions, such as sensorimotor integration (Buzsáki and Moser, 2013 [DOI](#)). However, increases in HPC theta power are also observed in hyperdopaminergic-like states, such as following the administration of ketamine or a dopaminergic agonist (Caixeta et al., 2013 [DOI](#); Thörn et al., 2022 [DOI](#)). Increases in theta power are reported in most studies in patients with schizophrenia (Newson and Thiagarajan, 2019 [DOI](#)) and have been correlated with poor cognitive performance (Curic et al., 2021 [DOI](#)). Thus, stronger HPC theta power does not necessarily reflect an optimal cognitive state. In fact, abnormal and increased theta power has been described as a consequence of ELS in different models and related to cognitive impairment (Kleen et al., 2011 [DOI](#); Velasquez et al., 2023 [DOI](#)).

Here we observed lower theta-gamma correlations in the PFC of ELS animals, suggesting a disconnection between theta power and local coordination of PFC activity. Similarly, the prolonged febrile model of ELS also presents decreased coordination of gamma activity in the hippocampal circuits (Kloc et al., 2023 [DOI](#)). In addition, we also observed a flatter PFC power spectrum in ELS animals compared to the normal tilt of CTRL, which has been discussed to indicate excitation/inhibition imbalance (Gao et al., 2017 [DOI](#)). These findings occurring during active behavior could explain the cognitive-behavioral alterations we observed, at least to some extent.

Desynchronization between limbic-frontal circuits is also reflected by reduced theta coherence during REM. Genetic and neurodevelopmental models of cognitive disorders exhibit a decrease in HPC-PFC theta synchrony correlated with poor spatial working memory performance and anxiety behaviors (Cunniff et al., 2020 [DOI](#); Sigurdsson, 2016 [DOI](#); Sigurdsson and Duvarci, 2016 [DOI](#)). Interestingly, increased HPC-PFC theta coherence was described in a model of repetitive ELS (Holmes et al., 2015 [DOI](#); Mouchati et al., 2019 [DOI](#)). Specifically, this enhanced coherence was observed during cognitive performance and associated with a mechanism of neuronal compensation (Kleen et al., 2011 [DOI](#); Niedecker et al., 2021 [DOI](#)). However, during REM sleep, these studies reported a decreased coherence and phase-locking value, indicating different synchronization dynamics depending on the brain state (Niedecker et al., 2021 [DOI](#)). Here, we observed impaired theta coherence only during REM sleep, which is the brain state with the expected highest coherence values, indicating that ELS animals could not reach or maintain this connectivity at the same maximum capacity as CTRL animals. The results from oscillatory dynamics in the HPC-PFC circuit indicate persistent state-dependent abnormalities in the temporal organization and interaction of local circuits that can contribute to long-term behavioral alterations and cognitive deficits. Further exploration should investigate HPC-PFC coordination during specific cognitive demands, such as working memory tasks, which we have shown to be impaired following ELS.

Our results also demonstrated that during ACT behavior, frontal-limbic oscillatory dynamics of ELS rats are unusually similar to REM sleep. Importantly, this finding could not be explained by altered locomotion or sleep architecture. This aberrant active state was more pronounced in the initial 12 h of recording, corresponding to the habituation to the novel environment. Dzirasa et al. (2006) [DOI](#) found similar results when recording hyperdopaminergic mice. The authors described a REM-like awake state in DAT-KO mice and wild-type mice injected with amphetamine. Similar to our results, the hyperdopaminergic wake state was marked by a significant increase in hippocampal theta oscillations. Additionally, treatment with antipsychotics reduced these REM-like hippocampal oscillations (Dzirasa et al., 2006 [DOI](#)). Our similar findings on a developmental model resulting in a psychotic-like phenotype support the relevance of intraindividual REM-wake similarity as a neural mechanism of psychosis. These results also advance previous studies, suggesting a shared involvement of dopamine in the impaired regulation of state-dependent global network dynamics in animal models of psychosis and epilepsy. Also interestingly, it is speculated that REM-wake similarity may represent a neural correlate of oneiric intrusion onto awake consciousness (Dzirasa et al., 2006 [DOI](#)), which, in the clinical context, could underlie potential phenomenological manifestations of sensorimotor disturbances of neurodevelopmental or psychiatric disorders.

In support of these dopaminergic alterations, we showed that ELS promotes apomorphine hypersensitivity and alterations in dopamine neurotransmission. Apomorphine sensitivity has been associated with reduced PPI, hyperlocomotion, and increased NAc dopamine release (Ellenbroek and Cools, 2002 [DOI](#)). Corroborating our data, neonatal SE promotes amphetamine sensitization and affects both dopaminergic and glutamatergic levels in prefrontal-striatal circuits (Lin et al., 2009 [DOI](#)). Importantly, alterations in dopamine signaling have been implicated in schizophrenia, ASD, and ADHD (Grace, 2016 [DOI](#); Kollins and Adcock, 2014 [DOI](#); Kosillo and Bateup, 2021 [DOI](#)). These results point to enhanced dopaminergic neurotransmission as a potential mechanism underlying the functional abnormalities observed in ELS rats.

In conclusion, we describe a comprehensive set of behavioral and neurobiological alterations following ELS that represent common patterns of dysfunctional circuit dynamics between developmental brain disturbances and multiple comorbidities associated. Although further studies should investigate causal links between neurobiological and behavioral alterations, our findings integrate a prominent theory about the pathogenesis of psychiatric symptoms that links stressful insults during development to increased HPC excitability, which would culminate in the hyperactivation of the mesolimbic system, resulting in behavioral dysfunctions (Gomes et al., 2020). Our results add depth to this theory by finding relationships among (1) neurodevelopmental insult, (2) neuroinflammation, (3) hyperexcitable HPC, (4) increased DA drive, (5) disrupted information flow in limbic-frontal circuits, and (6) psychiatric-relevant behaviors. Given the common pathophysiological and behavioral traits linking epileptogenesis and behavioral-cognitive comorbidities, we emphasize the experimental model's validity to investigate the neurobiological mechanisms of how an injury during development can lead to neuropsychiatric symptoms. Our study also demonstrates the potential of a multilevel strategy combining electrophysiology, behavior, immunohistochemistry, and neurochemistry in revealing integrated multidimensional mechanisms linking developmental impacts to epilepsy, cognition, and psychiatric symptoms.

Conflict of interest

The authors declare no competing financial interests.

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Reviewer #1 (Public Review):

Summary:

In this paper, Ruggiero, Leite, and colleagues assess the effects of early-life seizures on a large number of anatomical, physiological, behavioral, and neurochemical measures. They find that prolonged early-life seizures do not lead to obvious cell loss, but lead to astrogliosis, working memory deficits on the radial arm maze, increased startle response, decreased paired pulse inhibition, and increased hippocampal-PFC LTP. There was a U-shape relationship between LTP and cognitive deficits. There is increased theta power during the awake state in ELS animals but reduced PFC theta-gamma coupling and reduced theta HPC-PFC coherence. Theta coherence seems to be similar in ACT and REM states in ELS animals while in decreases in active relative REM in controls.

Strengths:

The main strength of the paper is the number of convergent techniques used to understand how hippocampal PFC neural dynamics and behavior change after early-life seizures. The sheer scale, breadth, and reach of the experiments are praiseworthy. It is clear that the paper is a major contribution to the field as far as understanding the impact of early-life seizures. The LTP findings are robust and provide an important avenue for future study. The experiments are performed carefully and the analysis is appropriate. The paper is well-written and the figures are clear.

Weaknesses:

The main weakness of the paper is the lack of causal manipulations to determine whether prevention or augmentation of any of the findings has any impact on behavior or cognition. Alternatively, if other manipulations would enhance working memory in ELS animals, it would be interesting to see the effects on any of these parameters measured in the paper. Also, I find the sections where correlations and dimensionality reduction techniques are used to compare all possible variables to each other less compelling than the rest of the paper (with the exception of the findings of U U-shaped relationship of cognition to LTP). In fact, I think these sections take away from the impact of the actual findings. Finally, the apomorphine section seemed to hang separately from the rest of the paper and did not seem to fit well.

Reviewer #2 (Public Review):

In this manuscript, the authors employ a multilevel approach to investigate the relationship between the hippocampal-prefrontal (HPC-PFC) network and long-term phenotypes resulting from early-life seizures (ELS). Their research begins by establishing an ELS rat model and conducting behavioral and neuropathological studies in adulthood. Subsequently, the manuscript delves into testing hypotheses concerning HPC-PFC network dysfunction. While the results are intriguing, my enthusiasm is tempered by concerns related to the logical flow, sample size, and the potential over-interpretation of results. Detailed comments are provided below:

Focus on Correlations: The manuscript primarily highlights correlations as the most significant findings. For instance, it demonstrates that ELS induces cognitive and sensorimotor impairments. However, it falls short of elucidating why these deficits are specifically linked to HPC-PFC synaptic plasticity/network. Furthermore, the manuscript mentions the involvement of other brain regions like the thalamus in the long-term outcomes of ELS based on immunohistochemistry data. This raises questions about the subjective nature and persuasiveness of the statistical studies presented.

Sample Size Concerns: The manuscript raises concerns about the adequacy of sample sizes in the study. The initial cohort for acute electrophysiology during ELS induction comprised only 5 rats, without a control group. Moreover, the behavioral tests involved 11 control and 14 ELS rats, but these same cohorts were used for over four different experiments. Subsequent electrophysiology and immunohistochemistry experiments used varying numbers of rats (7 to 11). Clarification is needed regarding whether these experiments utilized the same cohort and why the sample sizes differed. A power analysis should have been performed to justify sample sizes, especially given the complexity of the statistical analyses conducted.

Overinterpretation of HPC-PFC Network Dysfunction: The manuscript potentially overinterprets the role of HPC-PFC network dysfunction based on the results. Notably, cognitive deficits are described as subtle, with no evidence of learning deficits and only faint working memory impairments. However, sensorimotor deficits show promise. Consequently, it's essential to justify the emphasis on the HPC-PFC network as the primary mechanism underlying ELS-associated outcomes, especially when enhanced LTP is observed. Additionally, the manuscript seems to sideline neuropathological changes in the thalamus and the thalamus-to-PFC connection. The analysis lacks a direct assessment of the causal relationship between HPC-PFC dysfunction and ELS-associated outcomes, leaving a multitude of multilevel analyses yielding potential correlations without easily interpretable results.

Overall, while the manuscript presents intriguing findings related to the HPC-PFC network and ELS outcomes, it requires a more rigorous experimental design, a more coherent narrative linking results to hypotheses, and careful consideration of alternative interpretations based on the observed data. Addressing these concerns will enhance the manuscript's overall quality and impact.